

Racial Discrimination against Crime Victims: Evidence from Violent Crime Investigations

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Abstract

I develop and apply a new test for a consequential but overlooked form of racial discrimination: by police detectives against victims of crime. I test whether detectives display racial bias using a marginal outcome test derived from a model of police investigations. By performing the test on novel data on investigations of violent crime conducted by the Chicago Police Department over the last two decades, I find evidence of racial bias against Black victims: cases are 5.5 pp (6.1%) more likely to be rejected by the prosecution for insufficient evidence if the victim is Black rather than White. This is true for both fatal and non-fatal violent crimes, and holds true regardless the race of the alleged perpetrator. This disparity persists when focusing on court: violent crimes committed against Black victims are 7.6 pp (9.7%) less likely to result in a guilty ruling. These differences are consistent with the police being more tolerant of a failed deliverance of justice when the victim is Black. These results suggest that efforts to increase oversight and transparency in policing should also extend to police investigations and the broader differential treatment of crime victims.

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1 Introduction

The overly punitive approach of US police towards minority civilians suspected of a crime has been a major topic of discussion in recent years (CNN, 2014; National Academies of Sciences and Medicine, 2022). Research has shown that non-White civilians are more likely to be stopped and searched for contraband or weapons (Antonovics and Knight, 2009; Anwar and Fang, 2006), receive a traffic ticket (Goncalves and Mello, 2021), be arrested (Raphael and Rozo, 2019), and subject to the use of force (Hoekstra and Sloan, 2022; Lieberman, 2024) by the police.¹ Related work has documented that these forms of over-sanctioning by the police significantly erode trust in law enforcement institutions (Desmond et al., 2016; Ang et al., 2024).

Far less scrutinized is the alleged underperformance of the police in serving minority civilians when they are *victims* of a crime (Natapoff, 2006; Leovy, 2015). Minority residents of large US cities in fact report being discriminated against when victimized, arguing that police investigations are less thorough when the victim is not White. This is claimed to result in disproportionately many unsolved (i.e., not-*cleared*) crimes and lower levels of accountability for violence committed against these victims, with homicides cited as a leading example (Mueller and Baker, 2016; Tuerkheimer, 2017).² This differential treatment can have significant consequences. It can decrease the confidence in the police among minority residents and reduce their future willingness to cooperate, in addition to hindering crime deterrence in minority neighborhoods (Lowery et al., 2018a; Lowery et al., 2018b). Despite the importance of the issue, a rigorous examination of racial discrimination against crime victims is lacking.

In this paper, I fill this gap and test for racial discrimination against crime victims in the context of police investigations of violent crime. I develop a marginal outcome test for racial bias – as opposed to accurate statistical discrimination (Hull, 2021) – against crime victims, derived from a simple model of police investigations. Following Becker (1957), I leverage a post-treatment outcome (i.e., realized upon reaching the conclusion of what the police consider a successful investigation of the crime) to test for biased preferences among investigators. I apply this test to novel data

¹These disparities extend beyond interaction with the police, as suspected criminals are less likely to be released on bail (Arnold et al., 2018; Arnold et al., 2022) and more likely to be sentenced to prison and for longer (Abrams et al., 2012; Rehavi and Starr, 2014) if they are not White.

²Investigation of violent crime is not the only setting where racial discrimination against crime victims might appear. For example, non-White residents of large US cities have reported experiencing slow response times and a lack of attention and care from responding police officers after calling 911 (Sabino, 2024; Mahr and Annie, 2024).

from Chicago, which allows me to examine the behavior of an understudied decision-maker: the police detective. In this setting, the detective has a clear objective: to gather sufficient evidence to ensure a successful prosecution of the perpetrator(s), thereby delivering justice to the victims and their families. Intuitively, the detective will wait to submit a case for court proceedings until the expected success of her investigation has reached an acceptable level, given the evidence collected. Hence, all cases are submitted for prosecution exactly when this desired threshold of evidence, and consequently of success, is achieved.³

Empirically, I consider an investigation to be successful if the Cook County State Attorney’s Office (CCSAO) – which is the prosecutor of reference for the Chicago police – accepts the case for prosecution. Since this decision is based on the quantity and quality of the evidence presented by the police, it is informative about the detective work (Smith, 2019; Charles, 2022). Importantly, even when the prosecution rejects the case and no one is brought to court, the police can still claim to have solved the crime, ultimately stopping the investigation and classifying the case as an *exceptional clearance*, in contrast to the more standard *clearance by arrest and prosecution*.⁴

I then introduce the first testable definition of racial bias against crime victims in the context of police investigations handled by detectives, which differ in spirit from the one-time interactions typical of patrol officers. Specifically, bias originates from a lower acceptable threshold of expected success upon submission if the victim is not White. Simply put, cases are worked more poorly and are therefore less likely to enter the court process for the group subject to discrimination. These race-dependent thresholds can be micro-founded from a model of police investigation where the detective minimizes type 1 errors (i.e., submitting a case that fails) and type 2 errors (i.e., not submitting – yet – a potentially successful case). It can be shown that biased detectives hold a lower relative cost of type 1 error when the victim is not White.

I perform this marginal outcome test starting from novel data on 820,537 victims of violent crime and the related investigations handled by the CPD detectives from 2001 to 2023. Some striking descriptive facts emerge. Violent crimes are significantly less likely to be solved when committed against Black or Hispanic victims, and even less so when looking more narrowly at the probability of a crime being solved via arrest and prosecution. These gaps persist even after controlling for

³The model also allows for these thresholds to never be reached, which means that the case is never solved. This occurs if a case is suspended, formally or informally, and this decision can also be affected by the race of the victim.

⁴Another frequent reason for exceptional clearance is when the victim is not willing to cooperate and press charges. I discuss this instance more in detail in Section 2.

several features of the incident capturing the difficulty of the investigation, restricting to within-neighborhood comparisons, and accounting for the identity of the detective assigned to the case. These disparities, however, need not reflect racial bias: the detectives might be ensuring that all cases have the same likelihood of entering the court process, therefore accurately predicting the outcome of the next stage in the justice system. Any disparity in acceptance rates would instead point to racial bias. By restricting the focus to cases that are reviewed by the prosecution office, we can assess this through the lenses of an outcome test.⁵

Violent crimes investigated by CPD detectives are 6.4 pp less likely (-7%) to be accepted for prosecution upon submission when the victim is Black, consistent with racial bias defined as a higher tolerance for the failure to deliver justice to Black victims. I find modest evidence of bias against Hispanic victims as their cases are 2.2 pp less likely to be accepted for prosecution than cases of White victims. The main estimates are relatively unaffected by the inclusion of several characteristics of the originating incident and other demographics of the victim (-5.5 pp for Black victims and -1.5 pp for Hispanic victims). This suggests limited scope for bias originating from either other relevant attributes of the victim or the difficulty of the investigation. This also limits concerns about the logical validity of the marginal outcome test (Canay et al., 2023).⁶ Cases involving Black victims are less likely to be accepted upon submission across all types of violent crimes in the sample: homicides, robberies, sexual assaults, aggravated batteries and aggravated assaults.

Another relevant feature of the crime that need be considered, especially among cases that reach the submission stage, is the identity of the alleged criminal. It is possible that the desired thresholds are a function of the suspect’s race as well, and given that crime tends to occur within-race, the detectives’ behavior might be driven by the suspect’s characteristics, rather than the victim’s. The results, however, hold when controlling for the race – and sex and age – of the suspect: cases involving Black victims are less likely to be accepted regardless of the identity of the suspect.

I then tackle potential concerns about infra-marginality of the cases submitted by CPD (Ayres, 2002). If the probability of acceptance of the case does not evolve continuously over time, some

⁵As described in the body of the paper, only data from 2006 onward is available for the outcome test.

⁶Canay et al. (2023) shows that models of decision making underpinning outcome tests can be recast as Roy models. However, if the decision-maker follows a generalized Roy model, where unobserved characteristics enter both sides of the decision rule, we might conclude bias even if the decision-maker is unbiased or biased against the opposite group.

investigations might be considered concluded when their expected success has passed the desired threshold, which is particularly likely among very easy cases. If this is more likely to occur among cases of White victims, we could be incorrectly concluding bias among detectives. I address this by leveraging the time elapsed from when the crime was reported to when the case is reviewed by the prosecution. I specifically control for the duration of the investigation, under the assumption that the faster the case has been solved, the more likely the evidence is to lie above the threshold, and the controlling for this quantity would partial out any over-shooting of the thresholds. Nevertheless, I find that *even for cases submitted in the same amount of time*, a rejection is more likely if the victim is Black. Additionally, I drop cases solved very early, which are more likely to be infra-marginal, and the results are qualitatively unaffected.

The outcome of interest for the detective might also not be the *first* stage of the court process, i.e. the acceptance of the case, but the *last* one, i.e. a potential conviction. Using information from CCSAO, I document that cases involving Black victims are 7.4 pp less likely to reach arraignment and 7.6 pp less likely to result in a guilt ruling, even after accounting for the aforementioned controls. Detectives are also more tolerant of a higher risk of violent recidivism for crimes committed against Black victims, and to a minor extent for Hispanic victims. Moreover, consistent with the higher rates of rejections, crime involving Black victims are more likely to be classified as exceptional clearances due to prosecution denial rather than clearances achieved via arrest and prosecution. Regardless of the precise outcome guiding their behavior, detectives close relatively lower quality investigations for cases of Black victims, which is consistent with racial bias against them.

I next explore the drivers behind these results. First, it is possible that the detectives hand the same evidence for all victims to the prosecution, but CCSAO is more demanding for cases involving Black victims. This would mean that the detectives still tolerate a higher likelihood of rejection for cases of Black victims, and therefore still display bias, but do not *directly* set differential standards. Effective policy-making requires knowing which agent it can act upon. Second, racial bias can originate from two sources: biased *preferences* against minority victims (racial animus) ([Becker, 1957](#)) or biased *beliefs* held by the detectives about what makes an investigation successful upon completion (inaccurate statistical discrimination) ([Bohren et al., 2023](#)). Distinguishing between the two is important in devising the appropriate policy intervention aimed at correcting the detectives' practices.

I show that detectives play an active role in setting race-dependent thresholds by exploiting

relevant heterogeneity in the estimates. First, I find bias against Black victims throughout the entire time window, during which three different State Attorneys (SAs) have led the office, ranging from a police-friendly SA with a more orthodox approach to a more recent progressive-minded SA that often clashed with the police department (Conroy, 2001, Spielman, 2015, Grimm, 2022). The consistency of the estimates despite the radical change in prosecutorial approaches over the years suggests that the police preferences, at least to some extent, actively drive investigative behavior and, consequently, the differential rejection rates. Second, detective-level estimates of bias, obtained through a value-added approach, reveal substantial heterogeneity that cannot be explained by the tendencies of the ASAs who conduct the review, to whom cases are randomly assigned.

I examine the specific sources behind detectives' bias, which can be broadly classified as biased beliefs and/or biased preferences, by leveraging the detectives' characteristics (Arnold et al., 2018; Bohren et al., 2022). I find small and not significant interactions between the experience of the detective who clears the case and the race of the victim, suggesting that as detectives gain experience and, presumably, learn on the job, adjustments in beliefs do not change their differential submission standards. I then test for race concordance between victim and detective. If biased preferences were indeed behind the observed gaps, we would expect the rankings of the outcome by race of the victim, or at least the size of the gaps, to also depend on the race of the decision-maker (Anwar and Fang, 2006; Antonovics and Knight, 2009). These rankings, however, do not seem to vary with the race of the detective. I conclude that biased preferences are a more credible explanation than biased beliefs, but they are not a function of the race of the detective.

My findings suggest that transparency initiatives in policing, which focus on areas such as traffic stops and use of force, should also address police investigations and disparities in the treatment of crime victims more broadly. Additionally, using clearance rates as a metric for police performance is likely inadequate. Many cases considered solved involve arrests that never reach court proceedings, and they do so differentially by victim race.

Related Literature: I contribute to the literature on discrimination against minority individuals in the US justice system. A rich body of work in economics has studied the overly punitive tendency of the US justice system, and the police in particular, towards minority individuals. Most such work has focused on racial discrimination against individuals who are *suspected criminals*. Few papers have instead considered the treatment of crime *victims*. An exception is Alesina and

La Ferrara (2014), who tests whether Black defendants are more likely to be sentenced to death than White defendants by studying whether the ranking of error rates by defendant race changes with the race of the victim. Their main question, however, remains whether Black defendants are over-sanctioned. Moreover, they cannot identify marginal individuals, requiring strong assumptions on the distribution of the unobservables.⁷ Criminologists have devoted more attention to the treatment of victims, finding mixed results (Taylor et al., 2009; Petersen, 2017; Fallik, 2018). Most of these studies leverage data voluntarily reported by police agencies to the FBI (NIBRS) and compare the probability of clearance across victim races while controlling for factors that are thought to correlate with the difficulty of the case. Using these data and adopting a correlational approach raises identification concerns.⁸ Moreover, this literature has overlooked the identity of individual detectives and the link between investigations and subsequent stages of the justice system. I attempt to address these limitations. First, I define discrimination within a model of investigative behavior to guide identification. Second, I leverage data on individual detectives. Third, I use a court-related outcome realized upon crime solution to elicit decision-makers’ preferences.

I also contribute to the literature studying police behavior broadly defined. Most papers examining individual decisions made by the police have focused on the behavior of patrol officers. However, little is known about another crucial figure in law enforcement: the police detective. Detectives represent a relevant fraction of police employees and, importantly, significantly affect the probability of apprehension of criminals, determining the deterrent power of policing.⁹ Economists tangentially analyze the work of detectives when treating clearance rates as proxies for a police department’s performance, almost always as aggregates.¹⁰ To the best of my knowledge, this is the first paper that focuses on the individual decisions made by these agents and how they vary by the race of the victim involved in the assigned case.

Methodologically, I contribute to a large literature that develops methods to measure discrimination. A long tradition in economics has used outcome tests, which are based on differences in

⁷Another exception in economics is Bjerk (2022), which descriptively discusses differential patterns of clearance by victim and neighborhood, albeit in the context of a different research question.

⁸First, reporting to NIBRS is voluntary, poor and skewed towards smaller jurisdictions. This makes the results from these studies representative of medium- or small-sized cities, where policing and crime are radically different. Additionally, any null effect might be biased due to selection on consistently reporting to NIBRS. Moreover, these studies control for features that are either revealed only upon clearance or that directly affect investigative intensity, as noted by Cook and Mancik (2024).

⁹Of sworn officers employed by agencies with at least 100 officers, 27% reported working as investigators (Prince et al., 2021).

¹⁰For examples, see McCrary (2007); Mastrobuoni (2020); and Mastrococco and Ornaghi (2020).

post-decision outcomes across groups, to do so (Becker, 1957; Becker, 1993). A critical issue in this context is to identify individuals on the margin of being treated, as opposed to using group averages, which pool infra-marginal observations as well.¹¹ A common solution is to impose strong distributional assumptions on the unobservables of the individuals involved (Knowles et al., 2001; Anwar and Fang, 2006, Alesina and La Ferrara, 2014). Alternatively, recent work has instead leveraged quasi-random assignment to decision-makers to identify marginal individuals (Arnold et al., 2018; Dobbie et al., 2021). In the absence of quasi-random assignment, I circumvent the issue and identify racial bias by exploiting the dynamic nature of police investigations. Detective operations are concluded precisely when a desired threshold of evidence is reached, making all the observed solved cases on the margin of being closed.

Outline: The remainder of the paper proceeds as follows. Section 2 provides relevant background on how police clear a crime and the diverging paths a cleared case can take, especially in Chicago. Section 3 describes the main data used in the paper. Section 4 sketches a model of investigations and derives the marginal outcome test for racial bias in investigations. Section 5 presents the main results from the marginal outcomes test. Section 6 analyzes the sources of the observed bias. Section 7 discusses the policy implications and concludes.

2 Background

2.1 Crime Investigations in Chicago

Once a crime is reported to CPD, patrol officers are dispatched for the initial response. They conduct a preliminary investigation of the scene, provide assistance to the victim, interview witnesses, and apprehend any suspects who are still present (Chicago Police Department General Order G04-01). If the type of incident requires immediate notification per CPD Directives, the officers contact the respective Detective Area (or Police Area), which then dispatches detectives from the appropriate section (Chicago Police Department Special Order S04-21-01).¹² When on the scene, detectives from that detective area take over the investigation, processing physical evidence and interviewing

¹¹This is the so-called *infra-marginality* problem Ayres (2002).

¹²Each area had a Violent Crime Section and a Property Crime Section until 2020. In early 2020, CPD introduced a citywide Homicide Section and added homicide sections in each area.

persons having knowledge of the crime. If required, they are assisted by forensic technicians as well (Chicago Police Department General Order G04-02). Detective Areas are geographic partitions of the city where detectives are stationed. They encompass several districts and represent the level at which detectives are assigned to cases. For budgetary reasons, CPD reduced the number of these areas from five to three in March 2012, then restored them to five in April 2020 (Byrne, 2020). See Figure A.2 for the corresponding maps.

Detectives, like patrol officers, are assigned to one of three shifts. Case assignments tend to follow a rotation system, but only to some extent. In practice, assignments often deviate from quasi-random allocation within shifts, as detectives with particular knowledge of certain neighborhoods or types of incidents are more likely to be assigned those cases. The detectives dispatched to the scene are not necessarily those who conduct follow-up investigation as primary on the case.¹³ Table A.1 confirms this by showing that even conditional on area-by-time-by-crime type fixed effects, victims observe their cases being assigned to detectives belonging to the same race or ethnicity. This correlation persists even when controlling for other characteristics of the crime. Cases involving Black victims are, if anything, assigned to detectives with lower clearance rates than those assigned to cases of White victims.

2.2 How to Solve a Crime

According to FBI terminology, a solved crime corresponds to a “clearance”, and 2 main types of clearance are typically considered (FBI, 2010).¹⁴ First, a law enforcement agency can clear a crime by successfully arresting a suspect, which represents the vast majority of clearances. This occurs when the police have identified, arrested and turned over for prosecution at least one suspect. An alternative way of clearing a crime is by exceptional means. This occurs when the police have identified a suspect, gathered what they deem is sufficient evidence to support arrest and prosecution of the suspect, but encountered a circumstance beyond their control that prohibits them from doing so. Examples of these exceptional circumstances are victims’ refusal to cooperate,

¹³A review written in 2019 by the Police Executive Research Forum (PERF) on CPD’s investigative practices, per request of CPD and funded by DOJ, offers more details on homicide and shootings assignments. An instance of re-assignment is the “longstanding practice of midnight (or 1st watch) detectives transferring their cases to dayshift detectives. For homicide cases, these detectives typically respond to the scene, but then hand off the case to a dayshift detective for follow-up investigation” (PERF, 2019).

¹⁴The FBI defines these two types of clearance, and the circumstances leading up to them, mostly from a data collection perspective. In reality, law enforcement agencies sometimes deviate from these guidelines.

the prosecution’s refusal to press charges or the death of the identified offender.¹⁵ In essence, for the police to clear a crime, it often suffices to identify a credible suspect, regardless of whether they can apprehend and/or prosecute the individual.

Whether a crime is solved depends both on police behavior and incident-specific external factors, which determine the intrinsic difficulty of the investigation. For example, firearms leave less evidence than other weapons involving interpersonal contact, such as knives ([Asher, 2021](#)). Cooperation from witnesses (and/or victims if not a homicide), or the lack of thereof, is another influential factor that police report to strongly affect the likelihood of solving a crime ([Asher, 2021](#)). The actions police can take during a criminal investigation are numerous and multifaceted, ranging from the number and manner of interviews to the thoroughness of evidence collection and lab submissions, as well as persistence in follow-ups with witnesses, families, and victims.

2.3 At the Conclusion of the Investigation

While practices might vary by jurisdiction, whenever a suspect is found and is detained and the victim is willing to cooperate, the police typically decide whether to release her without charges or request that the prosecution press charges against her. In Cook County, the relevant jurisdiction for CPD, if the police opt for the latter, they contact the Felony Review Unit (FRU) at the CCSAO. This is typically a rotating group of Assistant State Attorneys (ASAs) who review the police’s evidence and decide whether to approve or reject the charges against the suspect in custody ([Cook County State’s Attorney, 2018](#), [Cook County State’s Attorney’s Office, 2024](#)). The police contact the FRU for felonies, while they can bypass the FRU for misdemeanors and drug charges (Chicago Police Department Special Order S04-21-01). The contacted ASA can decide whether to accept the charges based on the evidence presented, reject the charges or suggest that the investigation continue.¹⁶ Whenever charges are not accepted, the police release the suspect but can resubmit for charges at a later time. This is an instance when the police can classify the case as an exceptional clearance, ultimately stopping the investigation without any suspect going to court. The precise clearance classification is up to the police and does not necessarily perfectly match the result of the

¹⁵According to calculations made by [Kaplan \(2021\)](#) using 2019 NIBRS numbers from all agencies that reported data, approximately 50% of all exceptional clearances are due to a lack of cooperation from the victim, and 50% are due to the prosecution denial. These numbers mostly reflect trends for non-deadly crime, as homicides represent a very small share of total crime.

¹⁶From discussions with the CCSAO, continued investigation is suggested when there is a clear piece of evidence that is missing and is considered obtainable.

felony review, especially when multiple suspects are involved in the review.¹⁷ Whenever charges are accepted, the suspect(s) will be scheduled for a preliminary hearing before potentially appearing at arraignment, where they are formally charged, and going to trial and sentencing. If the case is accepted, it is typically classified as a clearance via arrest and prosecution, regardless of the outcome of the court process. Figure A.1 shows the arrest-to-court pipeline in Cook County, IL. The outcome test developed in this paper focuses for the most part on the first step, when after an arrest the police go to the CCSAO’s FRU.

From a legal standpoint, a discrepancy between police work and the prosecution’s opinion can arise because the standards for arrest and prosecution need not coincide. The law in fact requires that the totality of the evidence amounts to the level of *probable cause* for the police to be able to arrest and request charges on a suspect. However, for the prosecution to be able to convict the suspect(s), the threshold increases to *proof beyond a reasonable doubt*.¹⁸ It is therefore at the discretion of the police whether to gather evidence exceeding probable cause before submitting a case for prosecution or they deem probable cause to be sufficient. Analogously, the prosecution can accept cases that are not (yet) backed by evidence amounting to proof beyond reasonable doubt. However, given the incentive on their part to maximize conviction rates, it is in their best interest to reject a case whenever they deem the evidence insufficient. The FRU’s decision is therefore informative of the quality of the police work: acceptance of charges means that the CCSAO deemed the evidence to be good enough to seek a conviction, i.e., at the level of proof beyond reasonable doubt. Rejection can instead be interpreted as the CCSAO considering the evidence only at the level of probable cause. Note that even if charges are rejected, the police can consider the crime to be solved. In this case, the crime is classified as cleared by exceptional means. Upon rejection, the police decide whether to continue investigating and again seek charges or stop completely. The non-negligible rate of rejections by the CCSAO has been repeatedly highlighted

¹⁷Police agencies in the US handle exceptional clearances in different ways. Spohn and Tellis (2010) point out that a through interpretation of the FBI guidelines would entail cases being classified as exceptional clearances only when arrests are physically impossible. Many agencies, CPD included, adopt a laxer interpretation. Additionally, CPD and CCSAO seem to disagree on who governs the decision to exceptionally clear a crime. Police claim that it is the prosecutorial rejection that determines it, while CCSAO believes that the clearance classification is purely up to the police and ASAs solely opine on the quantity and quality of the evidence Riedel and Boulahanis, 2007.

¹⁸Probable cause means that, based on the existing evidence, a reasonable person would believe that an offense occurred and that it is probable that this suspect committed that offense. On the other hand, proof beyond reasonable doubt also requires that there is no reasonable doubt that the defendant committed the crime. For reference, see the US Supreme Court rulings in *Brinegar v. United States*, 338 U.S. 160 (1949) and *Maryland v. Pringle*, 540 U.S. 366 (2003). Probable cause and proof beyond reasonable doubt are not to be confused with *reasonable suspicion*, which is an even lower standard that is grounds for an investigative stop by law enforcement.

by the media as a problematic issue that can fuel a cycle of violence, especially in the context of murder investigations ([Smith, 2019](#); [Charles, 2022](#); [Grimm, 2022](#)).

3 Data and Descriptive Facts

3.1 Data

Through a series of FOIA requests, I obtained data on all victims of violent crime reported to the CPD from 2001 to mid-2024 and the related investigation. The data include information on both fatal violent crime (first-degree murder) and non-fatal violent crime (aggravated assault, aggravated battery, criminal sexual assault, robbery). I have information regarding the victim of the incident (race, ethnicity, gender, age), the incident itself (type of crime, weapon used, premises of incident, address, date and time, circumstances of the incident), all the detectives assigned as primary to the case, including their demographic characteristics (race, gender, date of appointment as police officer) and the date of assignment to each case, and the status of the investigation (whether the crime was cleared and the type of clearance, if cleared). The status allows me to distinguish between 3 major categories of clearances: clearances achieved via arrest and prosecution, exceptional clearances due to the victim’s refusal to cooperate and exceptional clearances for other reasons. The latter pools exceptional clearances due to prosecution denials and due to death of the offender. Since only a small of exceptional clearances occur due to the death of the offender, this third category mostly represents cases that are not prosecuted, as confirmed by communications with CPD ([Kaplan, 2021](#)).¹⁹ The status of the investigation does not always match the result of the felony review, among those cases that are reviewed. For this reason, the outcome test at the core of the paper will be based on the more precisely recorded result of felony review rather than the overall status of the investigation.

I drop all offenses that occurred in 2024 to allow some time for investigation. Unless specified, I keep only victims that are either White non-Hispanic (henceforth, White), Black or White Hispanic (henceforth, Hispanic), and I restrict the sample to cases that have ever been assigned to a detective.

²⁰ These two restrictions leave me with 751,999 victims. The dataset is kept at the victim-level

¹⁹For homicides, on the other hand, I can isolate cases cleared due to the death of the offender. This is because homicide-related information is recorded by CPD in a separate, more detailed database. Cases that are explicitly rejected for prosecution are instead labeled as “Bar to prosecute”.

²⁰Not all violent crimes have a primary detective assigned. These incidents are typically resolved on the scene by

since there is variation in demographics, mostly in age and gender, and features of the crime even within incidents.²¹ Although there can only be one primary detective at a time assigned to a case, they are sometimes replaced during the course of the investigation due to a variety of reasons (arbitrary decision of the supervisor, retirement, vacation, etc.). When using the full sample, I keep the first detective ever assigned to maintain consistency between solved and unsolved cases. When conducting the outcome test, which restricts to cases that are solved, I keep the detective that was on the case at the time of review of the case.

I also obtained data on all adults arrested by CPD employees from 2001 to 2024, which can be matched to the incident using a common identifier.²² This data provides information on the race, gender, age at time of the incident of the arrestee, time and location where the arrest took place, a list of all charges brought against them, with related type and class. I keep only arrestees who are either White, Hispanic or Black to match the filter imposed on crime victims. Importantly, the data contains information on felony review for each arrest that requires so. Namely, I observe the decision taken by the FRU and date and time of the review. This allows me to restrict the sample to those crimes that are linked to arrests that resulted in felony review. Unfortunately, information on felony review started to be systematically collected from 2006 onward. Therefore, the sample used for the outcome test is restricted to all adult arrests, involving a felony, made from 2006 to 2023 in connection to a violent crime committed anytime between 2001 and 2023. When using arrestee information, I expand the dataset to the victim-arrestee level and weight each observation by the product of total victims and total arrestees linked to the given incident.

Finally, I add court information using data obtained by the Chicago Data Collaborative from the Circuit Court of Cook County, which records information from arraignment to sentencing for each arrestee who reaches such stages. This data covers all court proceedings from 1984 to mid-2019, I further restrict the sample to felony arrests reviewed from 2006 to 2016 enough time for court proceedings to ensue.

the responding officers and do not require any follow up, or are more minor crimes that do not warrant a follow-up investigation (Special Order S04-21-01). Roughly 70% of the incidents without detectives assigned are cleared, and mostly via arrest and prosecution. The remaining 30% are for the most part assaults without injuries involved.

²¹More precisely, the dataset is at the victim-incident level. I cannot follow victims across incidents.

²²Data on juvenile arrests is not shared by CPD for privacy reasons.

3.2 Descriptive Statistics

Table 1 reports summary statistics for the full sample, separately for White, Black and Hispanic victims. Since homicides are vastly outnumbered by other types of violent crimes, but are also highly salient in the public minds, I keep them separate from non-fatal violent crimes. There is a large gap in the raw probability of clearance between White and minority victims of homicides. Cases of minority victims are also less likely to be solved via arrest and prosecution, even conditional on clearance. For non-fatal violent crime, in contrast, we observe a relatively small overall clearance gap which now seems to favor Black victims; however, this gap becomes more substantial and reflects the ranking observed for homicides if we consider clearances that occurred via the successful arrest and prosecution of a suspect. Consistently, refusal to cooperate is also most common among Black victims. This shows how the overall clearance rate can be a misleading measure of how often crimes get solved: only between slightly more than half of the victims whose cases are labeled as solved observe their offender going to court.

White victims of homicides are more likely to be female and older, and they represent a small share of total victims of homicide. Robberies are by far the most common of the non-fatal violent crimes, especially for White and Hispanic victims. Domestic violence is more prevalent among White victims if the incident ends up in a homicide, while the opposite is true for non-fatal violence. Firearm violence is more common among minority victims, especially for deadly acts of violence, where the share of firearm-related homicides exceeds 80%.

Table 2 shows the characteristics of the detective in my sample. Roughly two-thirds of the detectives working for CPD between 2001 and 2023 are White, Black and Hispanic detectives represent approximately 16% of the total each, and most of them are men. The average detective has 84% of her cases belonging to the same type of crime, but not even half of the total are hyper-specialized, which is consistent with anecdotal evidence (PERF, 2019).

Figure 1 shows the demographic distribution of neighborhoods, defined as Census tracts, in Chicago with the clearance rate. In only a very small share of Census tracts more than half violent crimes are solved and in the vast majority of neighborhoods less than one-third of these crimes are solved by arrest and prosecution. Figure 2 provides a more accurate picture of the relationship between race and the outcome of the investigations, by showing how clearance rates change as the share of minority residents in the Census tract increases. There is an overall negative relationship

when considering the overall clearance rate (Panel a), which however hides more nuanced trends. On one hand, if a crime is solved, the likelihood of this occurring via arrest and prosecution also decreases as the share of White residents decreases (Panel b). On the other hand, I document the opposite relationship for crimes solved due to the victim’s refusal to cooperate, conditional on the crime being solved (Panel c). All of these patterns become particularly stark in the top decile of the share of minority residents. Figure 2 illustrates further how the overall clearance rate is an imperfect measure of crime solution and accountability, especially when examining racial patterns.

3.3 Observational Comparisons

A natural question is whether differences in the probability of clearance persist once we compare very similar crimes committed against White and minority victims. This is an approach often adopted in the discrimination literature when investigating the disproportionate selection into an adverse treatment by a potentially biased decision-maker (Rehavi and Starr, 2014, Fryer Jr, 2019, Lieberman, 2024). The analysis in this setting, however, is complicated by the fact that we can identify four, rather two, major outcomes of an police investigation: 1) unsolved; 2) cleared via arrest and prosecution; 3) cleared by exceptional means due the victim’s refusal to cooperate; 4) other exceptional clearances, with the vast majority represented by cases when the prosecution refuses to press charges. Exceptional clearances are especially difficult to classify, as they are not obviously successful or failed investigations.

I still test for residual differences in the probability of clearance by race of the victim after controlling for several other features of the crime. I use 3 different definitions of clearance: *i*) overall clearance (cases belonging to categories 2), 3) and 4); *ii*) clearances *not* achieved due to the victim’s refusal too cooperate cases belonging to categories 2) and 4); *iii*) clearances achieved explicitly via arrest and prosecution (cases belonging to category 2)).

I estimate the following model:

$$D_i = \alpha + \beta^B Black_i + \beta^H Hispanic_i + \gamma' X_i + \delta_j + \delta_{ayc} + \varepsilon_i \quad (1)$$

where D_i is the observed clearance status of the case for victim i , which takes value 1 if the case has been solved. $Black_i$ and $Hispanic_i$ take value 1 if the victim is Black or Hispanic, respectively. The vector X_i includes a set of characteristics of the victim and the incident other than the race

of the victim that might correlate with it. These include: sex and age (binned in 6 categories) of the victim, whether the crime is gang-related, whether the crime is domestic violence-related, fixed effects for the weapon used in the crime, fixed effects for the premises where the crime occurred (e.g. street or residence), whether it was daylight or dark at the time of the crime, the day of the week when the crime occurred, the initial priority of the 911 call when the crime occurred, and how the crime was reported. I include fixed effects for the first detective assigned to the case (δ_j), and detective area-time-crime type fixed effects (δ_{ayc}). Time is defined as the specific month-year combination when the incident occurred, and there are five major types of crime (homicides, aggravated assaults, aggravated batteries, robberies, sexual assaults). I also progressively add controls for the neighborhood where the crime occurred. First, I include fixed effects for the quartiles of share of Black and Hispanic residents, separately, and the median income in the tract, logged to capture non-linearities.²³ Second, I replace these controls with tract fixed effects, which hold any time-invariant characteristics of the neighborhood fixed. I weight each observation by the inverse of the total number of victims involved in the incident. Standard errors are clustered at the detective area-time-crime type level.

As shown in Table 3, Black and Hispanic victims experience significantly lower clearance probabilities, regardless of the precise definition of clearance. Violent crimes involving Black (Hispanic) victims are roughly -3 pp (-2.4 pp) less likely to be solved when adopting a broad definition of clearance. For Black victims, these gaps increase in magnitude, to -4.2 pp, when not considering exceptional clearances due to the victim’s refusal to cooperate as solved. The difference in clearance probability slightly shrinks to -3.8 pp when narrowing the definition of clearance to cases solved via arrest and prosecution. For Hispanic victims, the estimates progressively decrease to -1.9 pp and then -1.3 pp. Disparities by characteristics of the neighborhood emerge as well, independently of the victim’s race. Clearance rates decrease once the share of Black residents exceeds 50%, while the gradient of the share of Hispanic residents is more monotonic. There is also evidence that median wealth in the Census tract positively affects the probability of clearance.

For these estimates to reflect discrimination, the included controls would have to perfectly capture the difficulty of the investigation, which can be considered in this setting the appropri-

²³I use the 5-years estimates from the 2010 American Community Survey. While the sample starts from 2001, estimates from the 2000 Census would be less representative of the racial makeup and income composition of the Chicago neighborhoods at the end of the 2010s and beginning of the 2020s.

ate measure of *qualification* to the treatment, with the treatment being clearance (Bohren et al., 2022).²⁴ Even if that was the case, it is not straightforward to classify the four major potential outcomes of an investigation into desirable and undesirable treatments. This is especially true for exceptional clearances due to the victim’s refusal to cooperate. The marginal outcome test developed in Section 4 addresses these challenges first, by restricting to cases that reach the prosecution’s office, whose victims are by definition willing to cooperate. In addition, difficulty is an issue for identification in the outcome test only insofar as it affects the thresholds of evidence that the detectives seek before concluding the investigation. Although difficulty almost certainly affects *whether* a case is solved, it is less obvious that it influences *how well* the case is closed, conditional on the case being solved, also given the legal standards provided by the law for this margin.²⁵

4 Model and Marginal Outcome Test

I next sketch a model of police investigations that allows me to develop a marginal outcome test for racial bias. Following Becker (1993), I leverage a post-decision (i.e., post-conclusion of the investigation), downstream outcome to elicit the agents’ preferences.²⁶ The model explicitly considers two key decision variables that can lead to a clearance. These are *i*) the time when the detective desires to clear and close the case; *ii*) the time when she would instead suspend altogether the investigation of a case that would therefore never be solved.²⁷

4.1 A Model of Police Investigations

Setup: The police receive case i , which is then assigned to an investigator. The investigator observes the race $R_i \in \{w, m\}$ of the victim, as well as other characteristics U_i that are in part unobserved to the econometrician. The investigator chooses how many days T_i to keep the case open and work on it before submitting it for clearance and, ultimately, for prosecution of the arrestee(s). She has the option of submitting the case on any day t following a decision rule that

²⁴Another issue is included-variable bias (Arnold et al., 2022).

²⁵As argued in Section 2, detectives need at least probable cause to make an arrest, but they are aware that the prosecution requires proof beyond reasonable doubt to win a case in court.

²⁶Post-treatment is slightly inaccurate for this setting since the outcome is realized at the time of clearance: once the case is ready to be closed, the outcome is also realized.

²⁷The test derived in this paper is similar to those in Anwar and Fang (2015) and Lodermeier (2023). They leverage the evolution of a relevant variable over time and use the (first) period when a decision is made by the relevant agent to identify marginal cases in the parole and eviction settings, respectively. Lodermeier (2023), in particular, does so without imposing strong parametric assumptions and is the closest in spirit to my approach.

tells her whether the case is ready for submission. Let $D_i(t) = 1$ denote the investigation being concluded and the case being submitted for prosecution – at time t .

Upon submission, the case can be successful or not, and this partly depends on how much time was spent on the case. As a measure of success, I use the DA’s decision to accept the case and proceed to prosecution. Let $Y_i^*(t) \in \{0, 1\}$ indicate the potential success when days $T_i = t$ are spent on the case before submission, and write the observed outcome as $Y_i = Y_i^*(T_i)$. The true probability of success upon submission of the case at the end of each period t is $p(t, r, u) = E[Y_i^*(t) | R_i = r, U_i = u, T_i = t]$. To simplify notation, expected success is denoted as $p(t)$. I assume that $p(t)$ is weakly increasing in t , all else fixed: the more time is spent on the case, the more evidence is collected, and the more successful the case is going to be in expected terms. I further assume that $p(0) = 0$ and that time is continuous.

The investigator also pays a subjective cost $c(R_i)$ of submitting a case with a given evidence level to court. I allow the cost to depend on race, with $c(w) \neq c(b)$ signifying racial discrimination. The lower $c(R_i)$ is, the more the detective is willing to submit at low values of $p(t)$, i.e., with worse evidence. The submission cost therefore identifies the desired evidentiary standard that has to be met before submission a victim of race R_i . The cost $c(R_i)$ can be further micro-founded by assuming that the detective minimizes the costs of type I errors (submitting potentially failing cases) and type II errors (not submitting – yet – potentially successful cases). $c(R_i)$ can be then recast as the relative cost of type I errors, i.e., submitting a potentially failing case, for a victim of race R_i .²⁸ Therefore, if she discriminates against minority victims, we expect $c(m) < c(w)$: failures are relatively less costly if the victim is not White.

Clearance: The investigator’s desire to clear, submit and close the case can then be represented as follows:

$$D_i(t) = \mathbf{1}[p(t) \geq c(R_i)] \quad (2)$$

There exists a period t_i^* satisfying the indifference condition $p(t_i^*) = c(R_i)$, at which point the investigator desires to submit and close the case. This is equivalent to the investigator having race-specific thresholds of minimum expected success that she is willing to tolerate before submitting, $\underline{p}(w)$ and $\underline{p}(m)$. If she has a distaste for investigating cases of non-White victims, we expect a lower probability of success for non-White cases filed on the margin than White cases filed on the margin,

²⁸See Appendix B.2 for the derivation.

i.e., $\underline{p}(m) < \underline{p}(w)$.

Note that these thresholds are assumed to depend on the race of the victim, but not on the other characteristics of the case, which jointly determine the difficulty of the investigation. These characteristics are assumed to affect the pace at which evidence accrues and how fast the case reaches the desired quality but not the desired thresholds of expected success themselves. In the empirical analysis, I relax this assumption and test whether any differential standards that I might find vary with observables other than race but potentially correlated with it. This assumption also preserves the logical validity of the marginal outcome test, a point stressed by [Canay et al. \(2023\)](#).

Suspension: Not all cases, however, will reach submission and clearance. The investigator can also suspend the investigation at any point in time even before submission if she considers the investigation not worth being kept open, by either formally or informally abandoning the case. This can be captured by the investigator also paying a subjective cost for actively investigating the case on day t , $l(R_i, U_i, t)$, which is potentially race-dependent and is strictly increasing over time. If she discriminates against non-White victims and dislikes investigating their cases, we expect $l(m, u, t) > l(w, u, t)$: she finds spending time on cases of non-White victims more costly than investigating cases of White victims.²⁹ If the investigation is suspended, the case will not be cleared or brought to court. Let $S_i(t) = 1$ denote the investigation being suspended at time t . The investigator will do so if the cost of investigation at time t exceeds the maximum effort that she is willing to exert on a case, $\bar{l}$:

$$S_i(t) = \mathbf{1}[l(R_i, U_i, t) \geq \bar{l}] \quad (3)$$

There is then a period t_i^0 such that the indifference condition along this margin holds, i.e., $l(R_i, U_i, t_i^0) = \bar{l}$. Equation 3 plays then the role of a (subjective) resource constraint forcing the detective into a corner solution, with the investigation not resulting in a clearance. While perhaps simplistic, Equation 3 is meant to capture the fact that the investigation will be suspended if it is considered unproductive and unlikely to reach the desired threshold of evidence/success for submission.

A case is therefore cleared and submitted if there is both enough evidence to do so, according to the investigator's subjective thresholds of success, as pinned down by $c(\cdot)$, and the investigation

²⁹While the suspension decision could be modeled in even greater detail, the decision is kept as simple as possible due to the lack of appropriate outcomes and data for an outcome test. Here, for example, I assume that the detectives know their cost of investigation. However, we can relax this assumption and make $l(R_i, U_i, t)$ an expected cost: $l(R_i, U_i, t) = E[l(t)|R_i, U_i]$.

was never abandoned. The realized time spent on the case is then either the time of suspension or the time of clearance, depending on whether the case was ever suspended, i.e., $T_i = S_i t_i^0 + (1 - S_i) t_i^*$. Clearance status will be equal to 1 only if the case was never suspended: $D_i = (1 - S_i)$. Success will always be equal to 0 if the case is never solved/submitted: $Y_i = D_i Y_i^*(T_i) = (1 - S_i) Y_i^*(T_i)$. Cases that are suspended will not arrive at the DA's office, and we therefore never observe their potential success.³⁰

A test for racial bias: In this model, there is room for discrimination along two margins: the investigator could both *i)* assign a lower chance of clearance to cases of Black victims by suspending potentially workable cases – discrimination on the extensive margin – and *ii)* set lower desired success rates even conditional on clearance/submission for cases of minority victims – discrimination on the intensive margin. Note that suspension and clearance here follow two decision processes that evolve in parallel.³¹

While the suspension decision does not involve a downstream potential outcome that is linked to the treatment, the submission decision does. Clearance is in fact determined once the expected downstream outcome Y_i , the successful submission for prosecution, is optimized relative to the costs. Moreover, all cleared cases are closed at the time of indifference, t_i^* . We can therefore perform a marginal outcome test for racial bias on the intensive margin by studying cases that are cleared and their success probabilities once they reach the SA's office. The observed disparity in success rates between cases involving non-White and White victims therefore identifies the difference in costs of submission, i.e., desired evidentiary standards, $c(r) - c(r')$. Since we defined this difference as racial bias, and cases are closed on the margin, we can state the following:

Proposition 1 *If $\Delta_t^Y \equiv E[Y_i \mid R_i = m, D_i = 1] - E[Y_i \mid R_i = w, D_i = 1] < 0$, the investigator displays racial bias on the intensive margin against non-White victims.*

We would therefore conclude that the police display bias against non-White victims if we estimate $\Delta^Y < 0$.³² For the proof of Proposition 1, see Appendix B.1.

³⁰Note that the suspension decision will affect which cases are submitted. Case i is not reviewed if $t_i^* > t_i^0$, making Δ^Y representative not of all investigations but only those that are brought to the end.

³¹This setting is close in spirit to use of force by the police: officers both decide whether to use force, potentially in a discriminatory way (Fryer Jr, 2019; Hoekstra and Sloan, 2022), and what *level* of force to use conditional on using force, also in a potentially in a discriminatory way (Lieberman, 2024).

³²Note that I follow Hull (2021) and allow bias to arise via non-race characteristics.

4.2 Empirical Approach

I next describe how I empirically estimate Δ^Y . I restrict the sample to cases that I know went through FRU and estimate the following baseline model:

$$Y_{ik} = \alpha + \beta^B \text{BlackVictim}_i + \beta^H \text{HispanicVictim}_i + \delta_j + \delta_{ayc} + \varepsilon_{ik} \quad (4)$$

where Y_{ik} is a dummy that takes value 1 if the case of victim i and arrestee k is accepted by the FRU. δ_j are detective fixed effects, and δ_{ayc} are detective area-month-year-crime type fixed effects. Detective area fixed effects are included because investigative operations (e.g., detective assignment, supervisor review) are conducted at this geographic level, and they can differ widely in their internal functioning and personnel. I interact these fixed effects with month-year fixed effects for when the case was reviewed, as the boundaries of the areas changed substantially over the years.³³ This also accounts for changes in the composition of the pool of ASAs reviewing the cases and the State Attorney herself. The coefficients of interest are β^B and β^H , which estimate total racial bias in closure decisions, against Black and Hispanic victims, respectively. I weight each observation by the inverse of the product of the total number of victims and total number of arrestees involved in the incident. The basic model is then augmented with additional fixed effects and controls to explore how the gap in success rates, if any, changes once we account for other factors that potentially correlate with race. Standard errors are clustered at the detective area-month-year-crime type level.

4.3 Sample for the Outcome Test

It is useful to remind the reader of the sample restrictions imposed for the outcome test. The sample is made of all adult arrests involving a felony that are made by CPD employees between 2006 and 2023 for violent crimes committed between 2001 and 2023. While also crimes that do not involve felonies can be dropped by the prosecution down the line, this can occur also because these are by definition minor crimes. Adopting the sample restrictions described here ensures that we consider only crimes that are reviewed immediately after the conclusion of the investigation and

³³For arrests made between 2006 and 2008, the arrest date rather than the review date is used since the review date is missing. This does not introduce virtually any measurement error since the felony review must occur by law within 48 hours of the arrest, as this is the maximum amount of time an individual can be kept in custody without being charged.

that are dropped due to lack of evidence and not for ethical concerns regarding the gravity of the crime.

As discussed in Section 2, the outcome of the felony review does not always align with the classification of the investigation, which is set at the case level and is common across all potential arrestees. Specifically, cases rejected by the FRU are not always recorded as exceptional clearances, and cases accepted by the FRU are not always recorded as clearances by arrest and prosecution. This mismatch arises for several reasons. First, when multiple arrestees are involved in a review and receive different decisions by the FRU, the ambiguity in the result gives the police considerable discretion in how to classify the investigation. This issue is particularly pronounced when arrestees are taken into custody and reviewed at different times. In addition, CPD does not offer detailed guidance on how to classify cases into clearance categories, allowing officers substantial discretion in departing from a shared standard, even when a single arrestee is presented to the FRU. Lastly, if a case is rejected, the police can still file misdemeanor charges against the suspect, which, however, entail much smaller penalties, typically different from a prison sentence. If the police still file misdemeanor charges, the officers might consider the case a successful clearance by arrest and prosecution. The outcome of the felony review is, therefore, the most accurate measurement of the success of an investigation for the k -th arrestee couple. Furthermore, this maps more straightforwardly to the court outcomes. The result of the felony review and the status of the investigations are still strongly related: the correlation between acceptance and clearance by arrest and prosecution is 0.4627 in the sample used for the outcome test, which rises to 0.4915 when restricting to crimes linked to only one arrestee.³⁴

5 Results

5.1 Main Results

Table 4 presents the results from the estimation of Equation 4. Column 1 shows a -9.1 pp (-3.5 pp) gap in acceptance rates between Black (Hispanic) and White victims of violent crime, which shrinks to -7.4 pp (-1.8 pp) when accounting for the different types of crime they experience (Column 2). When accounting for the area-time fixed effects and the identity of detective at review (Column 4), arrestees who commit a violent crime are 6.4 pp (2.2 pp) less likely to enter the court process if

³⁴Table A.3 reports summary statistics for the sample used for the outcome test.

the victim is Black (Hispanic) rather than White. This corresponds to a 7% gap for Black victims and a 2.4% gap for Hispanic victims.

The model upon which the test is based assumes that characteristics other than race can affect the expected rate of success of the case, and hence how fast evidence accrues and reaches the desired level, but they do not enter the cost function that pins down the required thresholds for submission. We can relax this assumption and test how the gap in success rates changes when we account for other observable characteristics of the victim, or the incident, that might correlate with race – and success. Racial bias could in fact originate *indirectly* from another characteristic that correlates with race and success (Bohren et al., 2022). In that case, by controlling for these characteristics, the estimates would shrink towards zero. However, we would still be in the presence of racial bias, despite it not *directly* originating from the race of the victim.

Table 5 performs this exercise, exploring how the baseline estimates, which are reported in Column 1, change once we progressively factor in other characteristics of the victim and the incident. Controlling for the sex and age of the victim leaves the estimates unaffected (Column 2), while including fixed effects for the quartiles of share of Black and Hispanic residents, separately, and the log of the median income in the tract decreases the differences for Black and Hispanic victims to -5.7 pp and -1.6 pp, respectively (Column 3). Interestingly, the coefficients relative to the racial makeup of the neighborhood are not significant, suggesting that racial bias originates from the identity of the single victim. On the other hand, a 1% increase in median per capita income in the tract is associated with a 1.4 pp increase in the probability of approval of the case, which is consistent with income discrimination as well, separately from racial bias. When replacing these neighborhood-level variables with census tracts fixed effects, the coefficients only marginally change (Column 4). The same is true when further controlling for the difficulty of the investigation (Column 5).

Another characteristic that is not the race of the victim but might correlate with it and with the success rates of different cases is the identity of the arrestee. Since violent crime tends to be committed within-race, for the race of the arrestee to explain the results and make the true gaps null, the CCSAO and/or the police would have to be particularly lenient towards minority arrestees, which seems unlikely given prior work on policing and prosecution behavior (Rehavi and Starr, 2014; Goncalves and Mello, 2021). Empirically, I control for race, gender and age of the arrestees and the coefficients remain virtually unchanged (Column 6). Contrary to what other papers have found, I find no evidence of over-punitiveness towards Black and Hispanic arrestees,

at least in the full sample used for the outcome test (Rehavi and Starr, 2014; Raphael and Rozo, 2019). After including all the aforementioned controls, the coefficients are -5.5 pp and -1.5 pp for Black and Hispanic victims, respectively. Note how these estimates have not dramatically changed with respect to Column 2 of Table 4, which simply accounted for the different types of crime these victims are exposed to. This is consistent with the assumption that focusing on the intensive margin, i.e. how well the case is solved, as opposed to the extensive margin, i.e. whether the case is solved, limits the confounding role of characteristics other than the victim’s race and facilitates the identification of the agent’s preferences.

I then explore whether the estimates vary by the specific type of violent crime by estimating a heterogeneous version of the main model, where I interact the race of victim and the arrestee with dummies for the four types of crime: aggravated assaults/batteries, sexual assaults, homicides and robberies. Figure 3 shows that the coefficients are relatively constant for Black victims with all but the one for aggravated assault/batteries being around -5 pp. The latter category is the one where Black victims experience the largest amount of bias (-8.6 pp). For Hispanic victims, the picture is more mixed, and they are discriminated against mostly if victims of violent crimes that have no sexual or financial motive, i.e. homicides and aggravated assault/batteries. Figure A.3 presents the estimates for the arrestees, obtained from the same regression. The main result that stands out is a remarkable tendency for over-punishment of Black and Hispanic individuals arrested in connection to a sexual assault and of Black suspects arrested for aggravated battery or assault .

5.2 Robustness

Next, I test the robustness of the main results to two important issues. I start by addressing potential concerns about infra-marginality. I then evaluate whether the detective are equalizing outcomes other than FRU acceptance.

5.2.1 Addressing Infra-Marginality

The model presented in Section 4 assumes that the expected acceptance of the case ($p(t)$) evolves continuously over time, and the cases are therefore submitted to FRU once this quantity equals the desired threshold of evidence. However, if $p(t)$ evolves discretely over time, some investigations might be considered concluded when their expected success has passed the desired threshold. This

would mean that the thresholds are confounded by infra-marginal cases (Ayres, 2002).³⁵ Such violation seems particularly likely among cases that are solved very quickly. Among these, the evidence could be so overwhelming that no further follow-up is required. Figure 4 plots the relationship between FRU acceptance and days elapsed from when the crime is reported to the review of the case, separately for the three racial groups considered. Acceptance is first residualized with respect to the main design controls, i.e. area-time-crime type fixed effects and detective fixed effects.³⁶ The range of days to review is truncated to six months to minimize sparsity. There is a stark gap between the residualized acceptance rates of White and Black victims that persists throughout the range of days to review. The relationship is more nuanced for Hispanic victims. Nevertheless, I take several steps to test whether infra-marginality is a concern for the main estimates.

First, I tackle this issue by controlling for the time elapsed between the offense and the review by CCSAO. This exercise lies on the assumption that over-shooting is a function of the length of the investigation, and once we control for it, the comparisons would be cleaner. I do so by including a third degree polynomial in days elapsed until review. Columns 1 and 2 of Table 6 present the results using the baseline regression (Column 4 in Table 4) and the most saturated model (Column 6 of Table 5). The estimates are virtually unchanged. We can conclude that even among cases solved in the same amount time, the suspected offenders are significantly less likely to face court proceedings and any degree of accountability if the victim is Black or Hispanic, rather than White.

Next, I drop cases that are solved relatively early, increasing the cutoff from one day, to two days, to a one week, again for the baseline model and the most saturated one. The coefficients for Black victims decrease slightly when dropping cases solved the same when the offense occurred (Columns 3 and 4) or the within next day (Columns 5 and 6), but increase again when keeping only cases that took at least one week of investigations to reach submission. Infra-marginality does not seem to be a concern for cases involving Black victims. On the other hand, violent crimes experienced by Hispanic victims have the same probability of acceptance when focusing on investigations that lasted at least one week. Notice that this might reflect cases of Hispanic victims being infra-marginal in the short run, or the desired thresholds varying with the duration of the investigation. It is possible that detectives' standards become less lax over time, i.e. $c(\cdot)$

³⁵Notice that in my model the detectives are not going to submit cases that have not reached the threshold, which means that expected success can “overshoot” but not “undershoot”.

³⁶Figure A.4 uses the raw acceptance rates.

is a function of time t as well, in a way to close the gap between Hispanic and White victims, potentially also due to the self-selection of the detectives into keeping a case open for longer.

Lastly, while in the main sample I do not find that minority arrestees are more likely to be subject to court proceedings, evidence of over-punitiveness towards them appears when dropping cases closed very early.

5.2.2 Other Outcomes

In the main model, I assume that the outcome the detective has in mind is the potential acceptance of the case by FRU. This is the first stage at the conclusion of the investigation, and therefore the least likely to be affected by other agents' bias and, most importantly, by exogenous events outside the control of the detective that might affect the relevant outcome. Nevertheless, it is possible that detectives account for these later stage decisions and events and optimize the potential guilt at the end of all court proceedings by the arrestee.

I test this by using data obtained by the Chicago Data Collaborative from the Circuit Court of Cook County on all court proceedings from 1984 to 2019. I match this information with the sample used so far for the outcome test using central booking numbers and restrict the sample further to arrests reviewed no later than 2016. This is done to allow sufficient time for court proceedings to ensue. In fact, if the length of court proceedings correlates both with race and the outcome, the estimates might be biased. The main outcome is whether the arrestee is associated with any guilty finding for that specific arrests. If the case never went beyond FRU, the outcome is by default a zero. Since this outcome is mechanically correlated with the number of charges brought against the individual arrestee, I further control for this variable in my models.

Table 7 reports the results. Even after controlling for all the available features of victim, arrestee and crime, suspects are 7.6 pp less likely to be found guilty if they commit violence against Black individuals than White individuals. The difference is sizeable, as it is 9.7% of the White victim mean. The magnitude of the coefficient for Hispanic victims matches that in Table 5 but it is not significant at conventional levels. Lastly, there is again neither evidence of over-punitiveness towards minority suspects, nor evidence of a relationship between neighborhood racial and income make-up and guilt in court.

Tables A.4 and A.5 consider alternative outcomes obtained from the court data. The former

uses as outcome whether the arrestee is found guilty of the top charge brought against them, which is typically the one corresponding to the i -th crime. The latter takes a step back in the court process and instead adopts as outcome whether the arrestee appears in court dataset in the first, which is roughly equivalent to the arrestee appearing at arraignment. This occurs if the case passes FRU and is considered viable by the arraignment prosecutor.³⁷ An arrestee is 7.4 pp (2.1 pp) less likely to appear at arraignment and 9.1 pp (2.3 pp) less likely to be found guilty of the top charge brought against them if they commit violence against a Black (Hispanic) victim rather than a White victim, with all these coefficients being significant.

The objective of police officers can also be framed as crime minimization. Detectives may therefore aim to equalize, across victim race, the probability that a suspect reoffends. The model in Section 4 can be reinterpreted with $p(t)$ representing the probability that the suspect k , who is potentially unknown at the beginning of the investigation, recidivates. Under this interpretation, $p(t)$ is now non-increasing in t : as more time is invested in the investigation, the likelihood of uncovering the culprit and apprehending them increases. Bias can now be understood as a higher tolerance for recidivism, i.e. a higher threshold applied to the discriminated group. A detective biased against group r perceives future crimes by suspects who initially offended against that group as less costly. These future crimes may be committed against the same victim, which is especially likely for batteries, sexual assault and domestic violence, or at least within the same demographic group, as prior research suggests (Becker, 2007).

I perform the same test using recidivism within one or three years of the arrest of suspect k in connection with the crime against victim i . Recidivism is defined based on arrests for either *i*) violent index crimes, or *ii*) violent index crimes involving a felony. These are serious offenses that impose higher costs on victims and are less subject to under-reporting. Table 8 reports the results. Suspects arrested for violent crimes against Black victims are 5 pp (2.8 pp) more likely to commit another serious violent crime within 3 years (1 year) than those arrested for violent crimes against White victims, even after controlling for differences in recidivism risk by suspect race. These effects are substantial, amounting to 17% (17.6%) of the White victim mean. When considering recidivism based violent crimes involving that *also* involve a felony, the differences slightly fall to 3.8 pp (13.8%) and 2 pp (13.9%) for the three- and one-year windows, respectively. For Hispanic

³⁷In Cook County, the ASA conducting the felony review is different from the ASA taking over the case after the initial review.

victims, the recidivism gap is 3.1 pp and 2.7 pp over three years, both statistically significant, but shrinks and becomes not significant over a one-year horizon.³⁸

Another outcome that can be used to perform the test is the status of the investigation, rather than the decision of the FRU or the court rulings. As discussed in Section 4, the status of the investigation can differ significantly from the FRU decision, especially when more than one arrestee is involved, and even more so if they are arrested and their cases reviewed at different times. It is useful to compare the results obtained using the FRU decision with those obtained with the status of the investigation, as the latter is the only information that is typically shared by police departments, when systematically collected. Table A.6 presents the results of the test using as an outcome an indicator that takes value 1 if the status of the investigation is clearance by arrest and prosecution, using the sample used so far in Column 2 and restricting to cases with only one arrestee involved in Column 4. For reference, results using the FRU decision as outcome are reported in Columns 1 and 3, respectively. The coefficients are roughly half the size when using the status of the investigation, which suggests that racial bias disparities would be under-estimated if the actual FRU decision were not available. While we cannot assess whether this is caused by intentional misclassifications or mistakes that correlate with race, these results call for improved data collection by police departments. The FBI has imposed police departments to use the NIBRS system, which entails a more detail data collection on the outcome of crime investigations, but this might still not be sufficient to have a clear picture of how many arrest enter the court process, as shown here.

5.3 Other Concerns

Table A.8 addresses other concerns. Columns 1 and 2 restrict the sample to cases involving only one victim or one arrestee, respectively. Column 3 combines these two filters. Column 4 does not weigh each observation by the inverse product of the number of victims and arrestees involved. Column 5 clusters the standard errors at the detective level rather than the area-time-crime type level. Column 6 uses the identity of the first detective assigned to the case rather than the one who is on the case at the time of the review. The results are qualitatively unaffected by these tests.

³⁸Table A.7 shows the results obtained using tract-level controls rather than census tract fixed effects.

6 Mechanisms

So far, I showed that cases involving violent crimes are less likely to be accepted by the prosecution, and the be linked to a guilty ruling in court, if the victim is Black, or Hispanic, rather than White. This is consistent with the detectives being more tolerant of a failure, i.e. lack of justice, is the victim is not White. I next dig deeper into the interpretation of these estimates and the drivers of this bias. First, the results so far have been framed as detectives' bias, but it is possible that the detectives hold the same standards for all victims to and CCSAO is disproportionately more demanding for cases involving Black victims. This would still be consistent with the detectives being more tolerant of a rejection for cases of Black victims, and therefore still display bias, but they would not be *directly* setting differential standards. Second, assuming that detectives play an active role, their bias can originate from two sources: biased *preferences* against minority victims (racial animus) (Becker, 1957) or biased *beliefs* regarding what the prosecution seeks (inaccurate statistical discrimination) (Bohren et al., 2023).

6.1 The Role of Detectives as Drivers of Bias

6.2 Variation Over Time

During the time window under analysis, 2006-2023, three different State Attorneys (SAs) have led the office: Richard Devine, a White man, at his third term, until December 2008; Anita Alvarez, a Hispanic woman, for two terms until December 2016; Kim Foxx, a Black woman, for two terms until December 2024. Besides the demographic differences, the first two are more orthodox police-friendly, though-on-crime prosecutors. Kim Foxx, on the other hand, campaigned on a reformist platform that championed de-prosecution and a rehabilitative approach, which made her often clash the police department (Conroy, 2001, Spielman, 2015, Grimm, 2022). If the estimates remain consistently negative throughout the time window, and especially under the more progressive Foxx, we could conclude that the police preferences, at least to some extent, actively drive investigative behavior and, consequently, the differential rejection rates.

I estimate a heterogeneous version of the main model, where I interact race with dummies for the specific term when case i was reviewed.³⁹ As shown in Figure 5, the estimates are negative

³⁹Notice that the only included term of Richard Devine is not complete, as it misses the first year, as well as Foxx's second term, which is missing the last year.

and significant for all terms among Black victims. Although we can reject that they are equal at the 10% level (the p-value for the null that they are all equal is 0.0703), we can safely reject that the gaps in rejection rates under Alvarez and Foxx are the same between Black and White victims (p-value=0.5089). The same holds for Hispanic victims, where all estimates are statistically equal (p-value=0.7173), even though none of them are significant. Figure A.5 presents the analogous estimates for Black and Hispanic arrestees, obtained from the same regression. No coefficient is significant and the magnitudes are relatively small, with the exception of Foxx second term, which is marked by a moderate tendency to over-punish Black arrestees.

6.2.1 Detective-level Estimates

If the detective thresholds matter for the acceptance of the case, and some of them systematically discriminate against minority victims, we should observe a considerable amount of between detective variation in acceptance gaps by victim race. Because cases are further randomly assigned to ASAs for felony review, the expected acceptance threshold applied by the ASA is independent of the detective assigned to the case. Consequently, variation in racial gaps in case acceptance across detectives reflects differences in detective case preparation and submission behavior rather than ASA-side bias. Moreover, how widespread racial bias in the justice system has become an important question in the literature (Goncalves and Mello, 2021).

For these reasons, I estimate detective-level racial bias by taking the most saturated model (Column 6 in Table 5), and interact the race of the victim with detective fixed effects, δ_j :

$$Y_{ik} = \alpha + \beta_j^B BlackVictim_i + \beta_j^H HispanicVictim_i + \gamma' X_{ik} + \delta_j + \delta_{ayc} + \varepsilon_{ik} \quad (5)$$

The ultimate goal is to estimate two meaningful quantities: the variation (standard deviation) in bias and the share of detectives who discriminate. The average detective-level bias is obviously also of interest, but we have already obtained a first estimate of this parameter in the main results.

These estimates are informative insofar as detectives close a sufficient number of cases for victims of all three demographic groups considered here. Therefore, I restrict the sample to detectives who close at least 10 cases for each group. An undesirable consequence of this sample restriction is that most of the sample is represented by robberies, namely 79%, while they constitute less than 50% of the full sample used for the outcome test. This is due to the fact that robbery detectives are

those that tend to work on, and therefore solve in absolute terms, disproportionately more cases. I then follow [Arnold et al. \(2022\)](#): they compute the mean and standard deviation of bias using empirical Bayes methods as in [Morris \(1983\)](#) and then the share of agents who discriminate, in this case corresponding to detectives with negative success gaps, using the posterior average effect calculation of [Bonhomme and Weidner \(2022\)](#). Standard errors for each of these quantities are obtained by bootstrapping shrunken estimates again derived following [Morris \(1983\)](#).

Figure 6 plots the distributions of the shrunk and un-shrunk detective-level effects, separately for Black and Hispanic victims, together with the aforementioned meaningful quantities and corresponding standard errors. Nearly two-thirds of the detectives in this sample discriminate against Black victims (SE=3.4 pp), while it is more of an even split for Hispanic victims (SE=3.2 pp). Importantly, there is wide variability in both types of bias, as standard deviations of 6.6 pp (SE=1.9 pp) and 6.3 pp (SE=2 pp) are estimated for Black and Hispanic victims. These numbers represent substantial variation, as they are 7.3% and 6.9% of the White outcome mean in this sample (.9084). This considerable amount of heterogeneity in the detectives’ acceptance gaps suggest that their attitudes matter, especially given that reviews are randomly assigned to ASAs.

6.3 Sources of Racial Bias: Leveraging the Detective’s Characteristics

The literature on discrimination has pointed out that what models of discrimination generally attribute to biased preferences against minority individuals (racial animus) ([Becker, 1957](#)) could instead originate from biased beliefs (inaccurate statistical discrimination). Here, these would be biased beliefs held by the detectives about what makes an investigation successful upon completion ([Bohren et al., 2023](#)). It is in fact difficult to distinguish between the two, and both correspond to forms of bias ([Arnold et al., 2018](#); [Arnold et al., 2022](#)). For policy reasons, however, it is important to understand whether we should intervene on beliefs and information to correct racial inequities or if some other levers have to be pulled, such as de-biasing training. It is important to note that racial animus in this setting may stem from several different sources, which cause $c(m)$ to be lower than $c(w)$. $c(r)$, in fact, includes personal racial attitudes of the detective, the organizational incentives from concluding a successful investigation for a victim of given race, the perceptions of the public. I attempt to distinguish between these two forms of bias by leveraging the characteristics of the detectives who clear the case, similarly to what [Arnold et al. \(2018\)](#) do with bail judges.

If inaccurate statistical discrimination explains the main results, cases of Black victims would be disproportionately more likely to be rejected by the ASAs because detectives hold inaccurate beliefs regarding evidentiary quality and/or standards rather than biased preferences. If this is the case, the more experienced the detective is, the smaller the gap in success rates due to correction to her beliefs. If we do not find any significant interaction between experience of the detective and the race of the victim, racial animus is a more likely explanation for the observed bias. Specifically, to test the role of – potentially inaccurate – beliefs, I estimate the following model:

$$\begin{aligned}
Y_{ik} = & \alpha + \beta^B BlackVictim_i + \beta^H HispanicVictim_i + \theta Experience_{jy} + \\
& + \theta^B BlackVictim_i \times Experience_{jy} + \theta^H HispanicVictim_i \times Experience_{jy} \\
& + \gamma' X_{ik} + \delta_j + \delta_{ayc} + \varepsilon_{ik}
\end{aligned} \tag{6}$$

Experience is calculated as the number of years between the review of a specific case and the year of first appointment as a police officer for the respective detective.⁴⁰ It is further de-measured so that the baseline coefficient for Black and Hispanic victims represents the gap in success rates for the detective with average experience.

Previous work on discrimination by law enforcement agents has tested for biased preferences by studying whether the rankings of the outcome, by race of civilian, are a function of the race of the decision-maker ([Anwar and Fang, 2006](#), [Antonovics and Knight, 2009](#), [West, 2018](#)). More recent work however has pointed out that this exercise is of limited informativeness ([Arnold et al., 2022](#)). I still perform this test by estimating the following model:

$$\begin{aligned}
Y_{ik} = & \alpha + \rho RaceMatch_{ij} + \beta^B BlackVictim_i + \beta^H HispanicVictim_i \\
& + \theta Experience_{jy} + \gamma' X_{ik} + \delta_j + \delta_{ayc} + \varepsilon_{ik}
\end{aligned} \tag{7}$$

where $RaceMatch_{ij}$ takes value 1 if victim and detective belong to the same racial group. I choose this approach rather than testing each pairwise combination to ease interpretation, similarly to [West \(2018\)](#).

Table 9 shows the results of these regressions. Gaining experience on the job does not seem to close the gaps in acceptance rates (Column 1). The interactions between Black and Hispanic

⁴⁰The year of first appointment as a detective is not available. Officers are promoted to the rank of detective years after their first appointment as patrol officers.

victims and experience of the detective are in fact negligible in magnitude and not significant. The baseline coefficient on experience, however, is statistically significant and it suggests that an additional year as a detective increases the probability that a case is accepted at submission by 1.2 pp. Hence, detectives seem to learn on the job how to get a case through FRU, this does not occur differentially by race of the victim once we account for all the available observables. In conclusion, we can reject that inaccurate beliefs on what is sought by CCSAO are an explanation for the main results.

When testing for race concordance (Column 2), the coefficient is not significant. This means that assigning a Black or Hispanic detective to a case of a Black or Hispanic victim, respectively, does not shrink the gaps in acceptance rates. This ultimately suggests that racial bias in this setting is more likely to originate from biased preferences than from biased beliefs ([Arnold et al., 2018](#); [Bohren et al., 2022](#)), but such differential standards are not a function of the detectives’ race. There are a few alternative explanations. First, detectives’ bias still stems from their personal attitudes but it is just not a function of their own race in this particular setting. Otherwise, detectives might be responding to incentives introduced by other agents. For example, upper management might differentially reward, or impose, acceptance rates of victims of different races. This is related to another potential explanation, which is the preferences of the public. Crimes committed against White victims might be more salient and a failure to bring them justice could be sanctioned more harshly.

7 Policy Implications and Conclusion

This paper provides the first evidence that the police discriminate by race when interacting with victims of crime. I do so by focusing on police investigations of violent crime. I develop a novel marginal outcome test for racial bias that does not require random assignment of cases to decision-makers and that exploits the dynamic nature of the setting. By leveraging the prosecution’s decision to accept or reject a case based on the evidence presented as the outcome of interest, I show that the detectives in the Chicago Police Department systematically close lower-quality investigations if the victim is Black relative to when the victim is White. For both fatal and non-fatal violent crime, Black victims are in fact less likely to ever see their assailant go to court. This is true regardless of the race of the alleged perpetrator and persists when looking at other court outcomes.

As detectives' experience does not seem to reduce these gaps, the estimates are unlikely driven by the detectives missing key information or having biased beliefs in general.

My results highlight several policy implications. First, calls for oversight and transparency in policing, which have been increasing in recent years, should also consider investigations, not only activities such as traffic stops and use of force, which have come under major scrutiny. While less salient in the public's mind than other policing activities when discussing policing reforms, the decisions of police investigators can also be biased, and they should be the object of external monitoring. Moreover, this paper documents that only a share of all crimes that are considered solved by the police actually end up entering the court process, due to the prosecution deeming the evidence presented by the police to be insufficient. The issue becomes even more sizable if we consider non-fatal violent crime and the remarkable share of clearances achieved due to the victim's refusal to cooperate. For these reasons, the overall clearance rate seems an inadequate metric to measure the performance of a police department, a concern echoed by [Cook and Mancik \(2024\)](#) and [Baughman \(2020\)](#).

References

- Abrams, David S, Marianne Bertrand, and Sendhil Mullainathan, “Do judges vary in their treatment of race?,” *The Journal of Legal Studies*, 2012, 41 (2), 347–383.
- Alesina, Alberto and Eliana La Ferrara, “A test of racial bias in capital sentencing,” *American Economic Review*, 2014, 104 (11), 3397–3433.
- Ang, Desmond, Panka Bencsik, Jesse M Bruhn, and Ellora Derenoncourt, “Community Engagement with Law Enforcement after High-profile Acts of Police Violence,” Technical Report, National Bureau of Economic Research 2024.
- Antonovics, Kate and Brian G Knight, “A new look at racial profiling: Evidence from the Boston Police Department,” *The Review of Economics and Statistics*, 2009, 91 (1), 163–177.
- Anwar, Shamena and Hanming Fang, “An alternative test of racial prejudice in motor vehicle searches: Theory and evidence,” *American Economic Review*, 2006, 96 (1), 127–151.
- and —, “Testing for racial prejudice in the parole board release process: Theory and evidence,” *The Journal of Legal Studies*, 2015, 44 (1), 1–37.
- Arnold, David, Will Dobbie, and Crystal S Yang, “Racial bias in bail decisions,” *The Quarterly Journal of Economics*, 2018, 133 (4), 1885–1932.
- , —, and Peter Hull, “Measuring Racial Discrimination in Bail Decisions,” *American Economic Review*, 2022, 112 (9).
- Asher, J, “What improves the chances of solving a murder,” *The New York Times*, 2021.
- Ayres, Ian, “Outcome tests of racial disparities in police practices,” *Justice research and Policy*, 2002, 4 (1-2), 131–142.
- Baughman, Shima Baradaran, “How effective are police? The problem of clearance rates and criminal accountability,” *Ala. L. Rev.*, 2020, 72, 47.
- Becker, Gary S, *The economics of discrimination*, University of Chicago press, 1957.
- , “Nobel lecture: The economic way of looking at behavior,” *Journal of political economy*, 1993, 101 (3), 385–409.
- Becker, Sarah, “Race and violent offender “propensity”: Does the intraracial nature of violent crime persist on the local level?,” *Justice Research and Policy*, 2007, 9 (2), 53–86.
- Bjerk, David, “Does greater police funding help catch more murderers?,” *Journal of Empirical Legal Studies*, 2022, 19 (3), 528–559.
- Bohren, J Aislinn, Kareem Haggag, Alex Imas, and Devin G Pope, “Inaccurate statistical discrimination: An identification problem,” *Review of Economics and Statistics*, 2023, pp. 1–45.
- , Peter Hull, and Alex Imas, “Systemic discrimination: Theory and measurement,” Technical Report, National Bureau of Economic Research 2022.

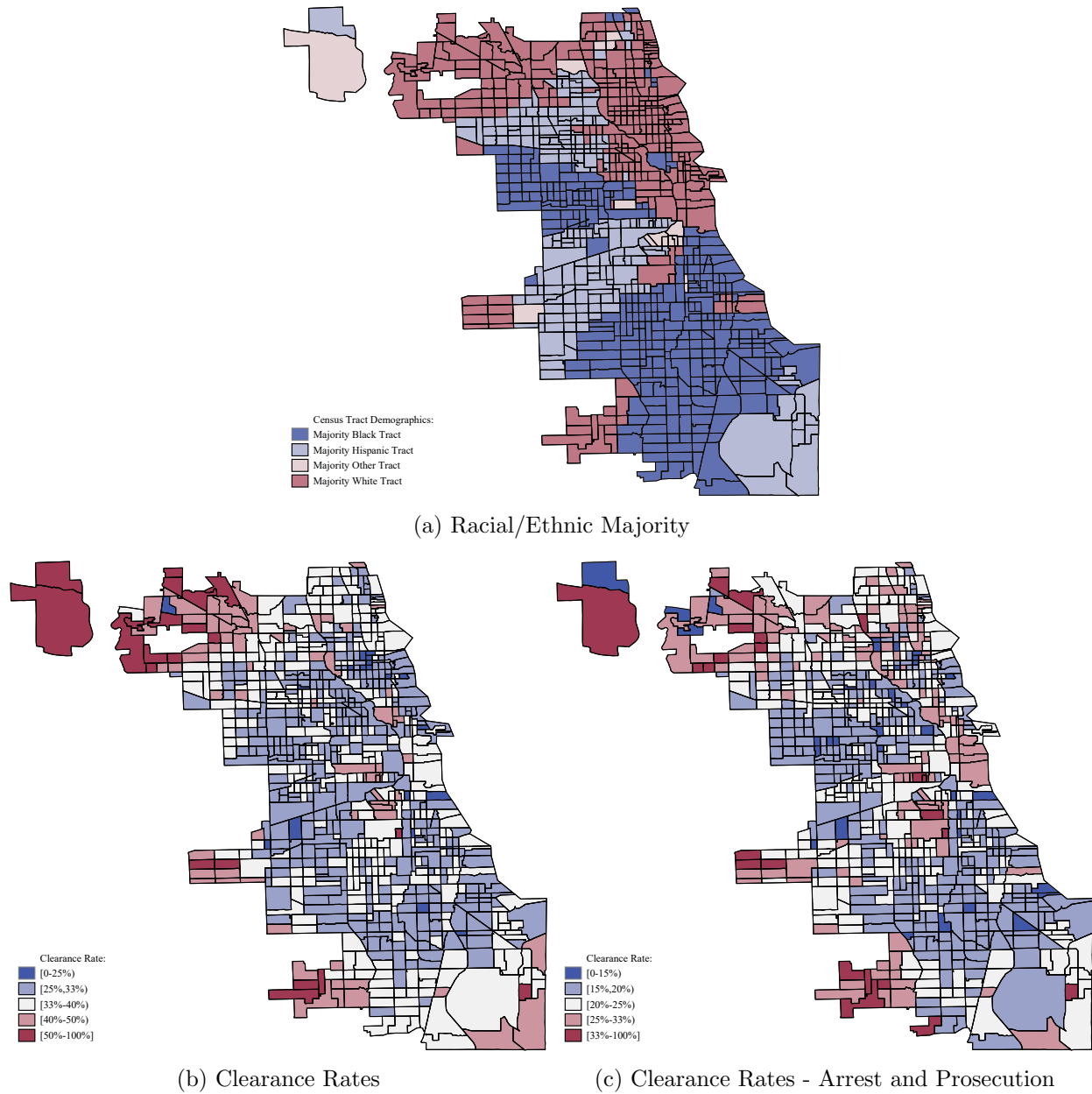
- Bonhomme, Stéphane and Martin Weidner**, “Posterior average effects,” *Journal of Business & Economic Statistics*, 2022, 40 (4), 1849–1862.
- Byrne, John**, “Mayor Lori Lightfoot reopens shuttered Chicago police detective areas,” *Chicago Tribune*, 2020.
- Canay, Ivan A, Magne Mogstad, and Jack Mountjoy**, “On the use of outcome tests for detecting bias in decision making,” *Review of Economic Studies*, 2023, p. rdad082.
- Charles, Sam**, “As CPD murder clearances rise, prosecutors more often decline to bring charges, data show,” *WGN9 TV*, 2022.
- CNN**, “Poll: Racial divide in views on justice system, police,” 2014. Accessed: 2024-10-21.
- Conroy, John**, “A Hell of a Deal. Solving Chicago’s murders could prevent more,” *Chicago Reader*, 2001.
- Cook County State’s Attorney**, “Felony Review Manual,” 2018.
- Cook County State’s Attorney’s Office**, “Criminal Prosecutions Bureau,” 2024.
- Cook, Philip J and Ashley Mancik**, “The sixty-year trajectory of homicide clearance rates: Toward a better understanding of the great decline,” *Annual Review of Criminology*, 2024, 7, 59–83.
- Desmond, Matthew, Andrew V Papachristos, and David S Kirk**, “Police violence and citizen crime reporting in the black community,” *American sociological review*, 2016, 81 (5), 857–876.
- Dobbie, Will, Andres Liberman, Daniel Paravisini, and Vikram Pathania**, “Measuring bias in consumer lending,” *The Review of Economic Studies*, 2021, 88 (6), 2799–2832.
- Fallik, Seth Wyatt**, “Detective effort: what contributes to arrests during retrospective criminal investigations?,” *Policing and Society*, 2018, 28 (9), 1084–1104.
- FBI**, “Offenses Cleared (<https://ucr.fbi.gov/crime-in-the-u.s/2010/crime-in-the-u.s.-2010/clearances>),” 2010.
- Goncalves, Felipe and Steven Mello**, “A few bad apples? Racial bias in policing,” *American Economic Review*, 2021, 111 (5), 1406–1441.
- Grimm, Andy**, “Half of murder cases considered ‘solved’ by Chicago police in 2021 didn’t lead to charges,” *Chicago Sun-Times*, 2022.
- Hoekstra, Mark and CarlyWill Sloan**, “Does race matter for police use of force? Evidence from 911 calls,” *American economic review*, 2022, 112 (3), 827–860.
- Hull, Peter**, “What marginal outcome tests can tell us about racially biased decision-making,” Technical Report, National Bureau of Economic Research 2021.
- Jr, Roland G Fryer**, “An empirical analysis of racial differences in police use of force,” *Journal of Political Economy*, 2019, 127 (3), 1210–1261.

- Kaplan, Jacob**, *National Incident-Based Reporting System (NIBRS) Data: A Practitioner's Guide* 2021.
- Knowles, John, Nicola Persico, and Petra Todd**, "Racial bias in motor vehicle searches: Theory and evidence," *Journal of political economy*, 2001, 109 (1), 203–229.
- Leovy, Jill**, *Ghettoside: A true story of murder in America*, One World, 2015.
- Lieberman, Carl**, "Variation in racial disparities in police use of force," *Journal of Urban Economics*, 2024, 141, 103602.
- Lodermeier, Alison**, "Racial Discrimination in Eviction Filing," 2023.
- Lowery, Wesley, Kimbriell Kelly, Ted Mellnik, and Steven Rich**, "Where murders go unsolved: A Post investigation," *The Washington Post*, 2018.
- , **Steven Rich, and Georges Salwan**, "As police struggle to solve homicides, Baltimore residents see an open season for killing," *The Washington Post*, 2018.
- Mahr, Joe and Sweeney Annie**, "Many 911 calls deserve an 'immediate' police response. But in thousands of cases, officers didn't arrive for more than an hour," *Chicago Tribune*, 2024.
- Mastrobuoni, Giovanni**, "Crime is terribly revealing: Information technology and police productivity," *The Review of Economic Studies*, 2020, 87 (6), 2727–2753.
- Mastrorocco, Nicola and Arianna Ornaghi**, "Who watches the watchmen? Local news and police behavior in the United States," 2020.
- McCrary, Justin**, "The effect of court-ordered hiring quotas on the composition and quality of police," *American Economic Review*, 2007, 97 (1), 318–353.
- Morris, Carl N**, "Parametric empirical Bayes inference: theory and applications," *Journal of the American statistical Association*, 1983, 78 (381), 47–55.
- Mueller, Benjamin and A Baker**, "A mother is shot dead on a playground, and a sea of witnesses goes silent," *The New York Times*, 2016.
- Natapoff, Alexandra**, "Underenforcement," *Fordham L. Rev.*, 2006, 75, 1715.
- of Sciences, Engineering National Academies and Medicine**, "To reduce racial inequality in the criminal justice system, government should explore ways to reduce police stops, detention, and long sentences," 2022.
- PERF**, "Review of the Chicago Police Department's Homicide Investigation Process," 2019.
- Petersen, Nick**, "Neighbourhood context and unsolved murders: The social ecology of homicide investigations," *Policing and society*, 2017, 27 (4), 372–392.
- Prince, Heather, Cynthia Lum, and Christopher S Koper**, "Effective police investigative practices: an evidence-assessment of the research," *Policing: An International Journal*, 2021, 44 (4), 683–707.

- Raphael, Steven and Sandra V Rozo**, “Racial disparities in the acquisition of juvenile arrest records,” *Journal of Labor Economics*, 2019, 37 (S1), S125–S159.
- Rehavi, M Marit and Sonja B Starr**, “Racial disparity in federal criminal sentences,” *Journal of Political Economy*, 2014, 122 (6), 1320–1354.
- Riedel, Marc and John G Boulahanis**, “Homicides exceptionally cleared and cleared by arrest: An exploratory study of police/prosecutor outcomes,” *Homicide Studies*, 2007, 11 (2), 151–164.
- Sabino, Pascal**, “911 calls on South, West Sides ignored while ‘rapid response’ cops make traffic stops instead,” *Injustice Watch*, 2024.
- Smith, Patrick**, “Even When Chicago Police Solve A Murder, Justice Is Not Always Served,” *WBEZ Chicago*, 2019.
- Spielman, Fran**, “Alvarez accused of deliberately filing wrong charge against police detective,” *Chicago Sun Times*, 2015.
- Spohn, Cassia and Katharine Tellis**, “Justice denied: The exceptional clearance of rape cases in Los Angeles,” *Alb. L. Rev.*, 2010, 74, 1379.
- Taylor, Terrance J, David Holleran, and Volkan Topalli**, “Racial bias in case processing: does victim race affect police clearance of violent crime incidents?,” *Justice Quarterly*, 2009, 26 (3), 562–591.
- Tuerkheimer, Deborah**, “Criminal justice and the mattering of lives,” *Mich. L. Rev.*, 2017, 116, 1145.
- West, Jeremy**, “Racial bias in police investigations,” Retrieved from University of California, Santa Cruz website: <https://people.ucsc.edu/jwest1/articles/WestRacialBiasPolice.pdf>, 2018.

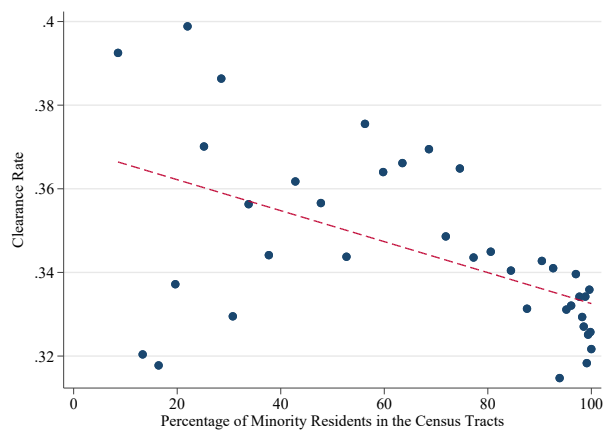
Figures and Tables

Figure 1: Racial/Ethnic Majority and Clearance Rates by Census Tract

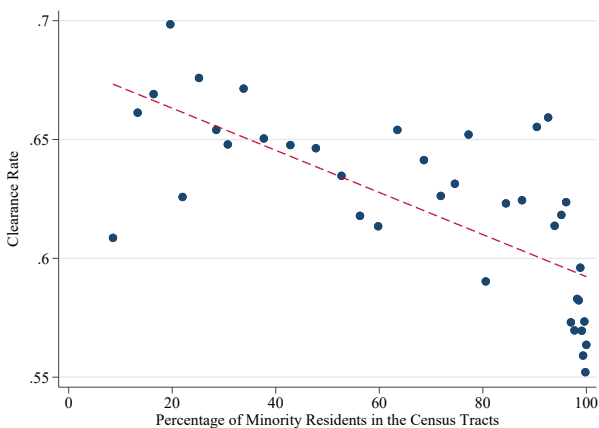


Notes: This Figure shows the demographic composition and the clearance rate over the time period 2001-2023 in Chicago Census tracts. Demographic information comes from the 2010 ACS.

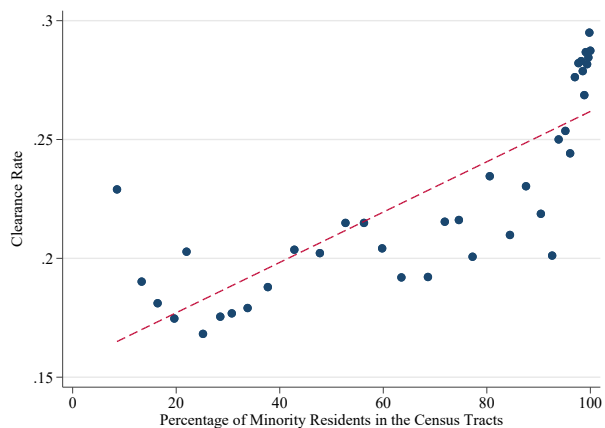
Figure 2: Relationship Between Clearance Rates and Percent Minority Residents in the Census Tract



(a) Clearance Rate



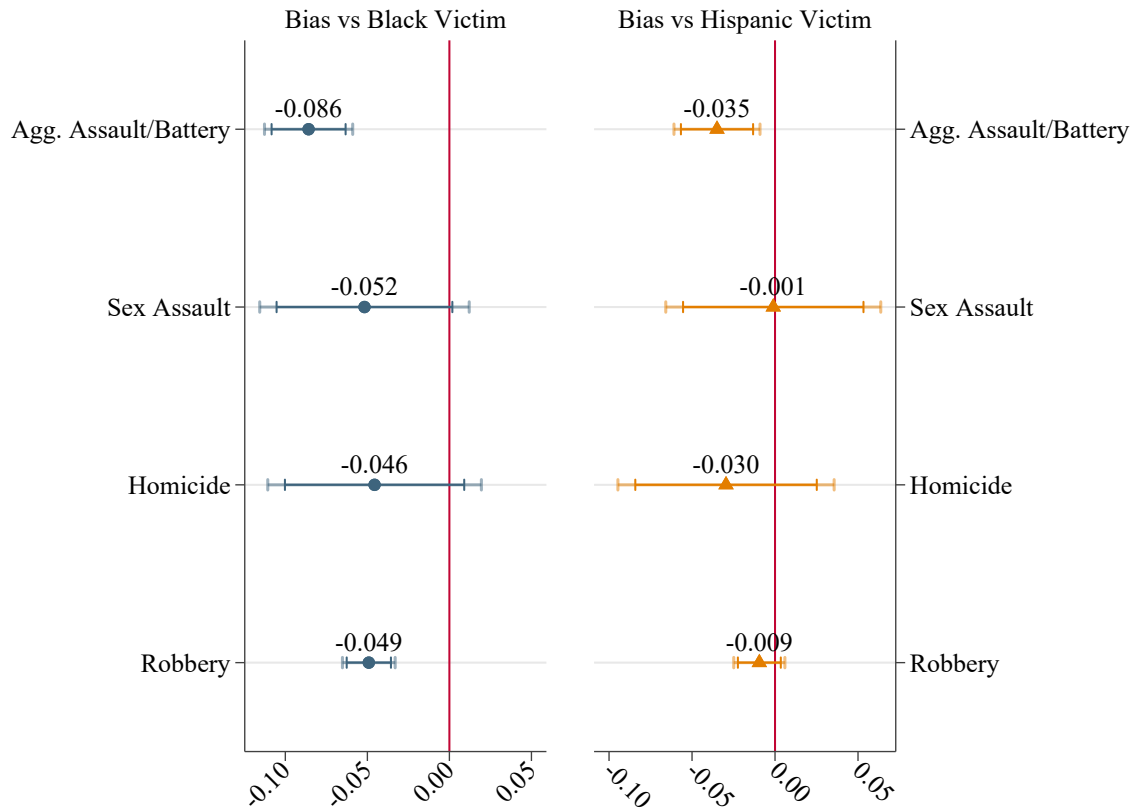
(b) Arrest and Prosecution | Clearance



(c) Victim's Refusal to Cooperate | Clearance

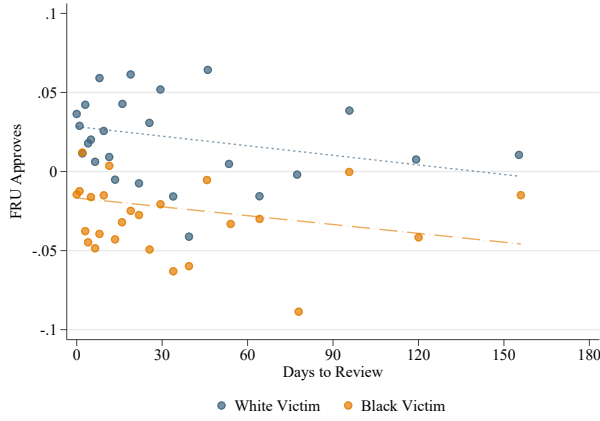
Notes: This Figures shows a binned scatterplot of the relationship between clearance rates and share of minority residents in Chicago Census tracts, with a linear fit. Panel a) uses the overall clearance rate, while panels b) and c) display the conditional clearances rates by arrest and prosecution and by exceptional means due to the victim's refusal to cooperate.

Figure 3: Heterogeneity in Racial Bias by Type of Violent Crime

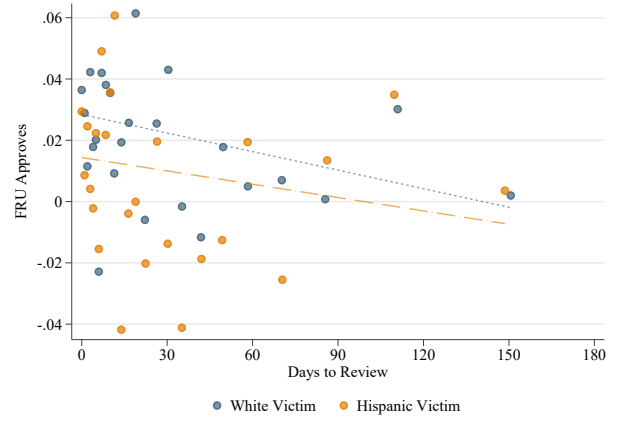


Notes: This figure plots the coefficient from a heterogeneous version of the model used in Column 6 of 5 by type of violent crime. Race dummies for victims and arrestees are interacted with fixed effects for the type of violent crime. This figure shows the coefficients on Black and Hispanic victims, separately.

Figure 4: Relationship between Residualized Acceptance and Days to Clearance



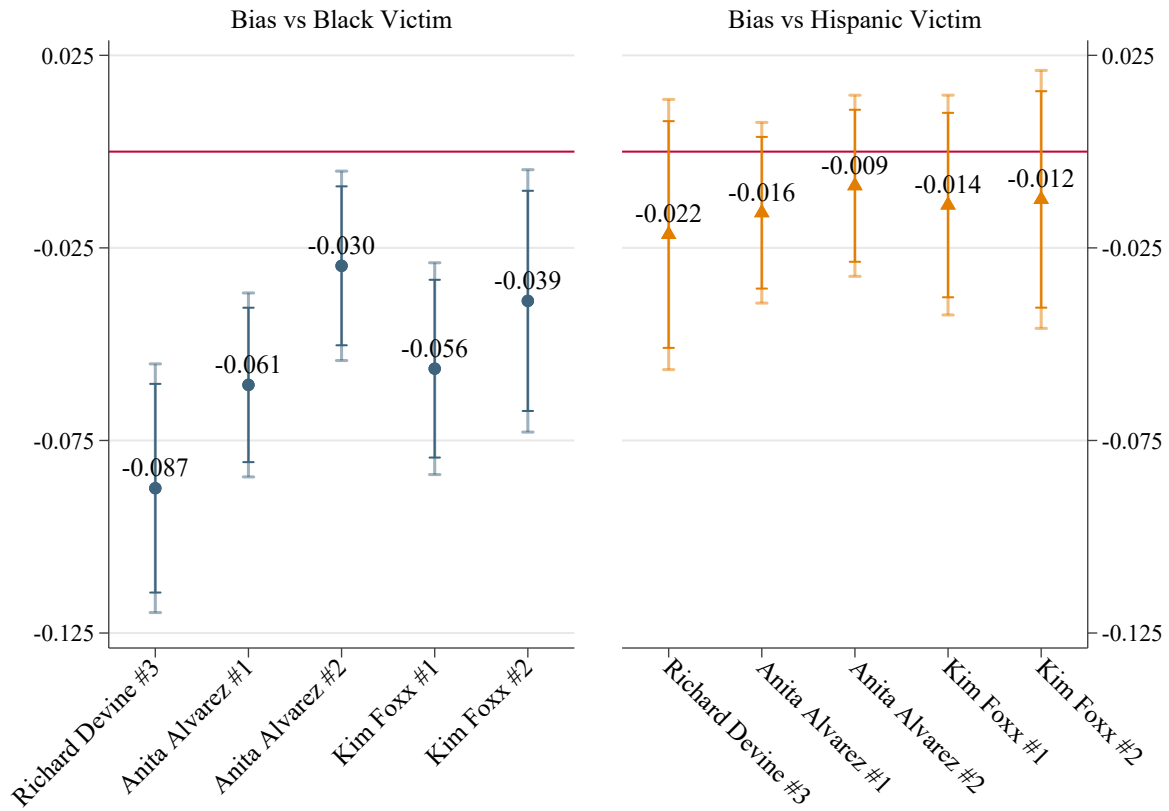
(a) White Victim vs. Black Victim



(b) White Victim vs. Hispanic Victim

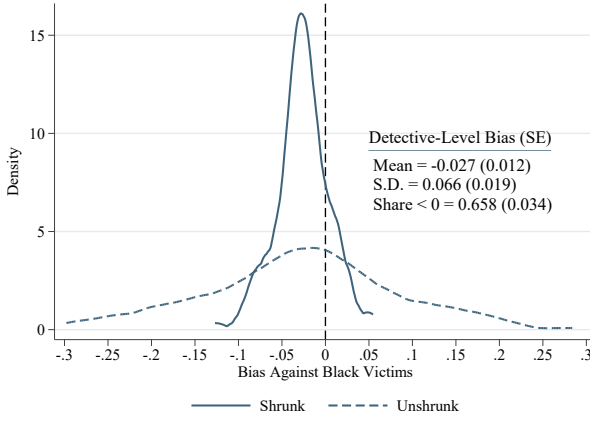
Notes: This figure plots a bin scatter of the relationship between acceptance by FRU and days elapsed from report of the crime to review, by race of the victim. Acceptance is first residualized with respect to area-time-crime type fixed effects and detective fixed effects. The range of days to review is truncated at 180 days.

Figure 5: Heterogeneity in Racial Bias by SA's Term

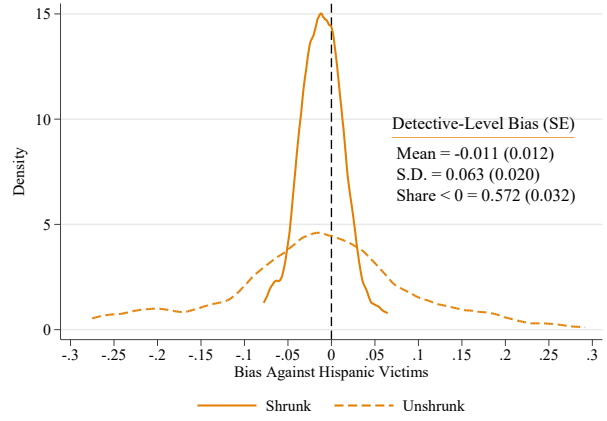


Notes: This figure plots the coefficient from a heterogeneous version of the model used in Column 6 of 5 by term of the SA leading the CCSAO. Race dummies for victims and arrestees are interacted with fixed effects for the term of the SA leading the CCSAO. This figure shows the coefficients on Black and Hispanic victims, separately.

Figure 6: Detective-level Estimates of Bias



(a) White Victim vs. Black Victim



(b) White Victim vs. Hispanic Victim

Notes: This figure plots the distribution of detective-level bias, shrunk and unshrunk. Detective-level estimates of bias are obtained from Model 5, specifically as the interaction between dummies for the race of the victim with detective fixed effects. The fractions of discriminators are computed from these estimates as posterior average effects following [Arnold et al. \(2022\)](#), who build from [Bonhomme and Weidner \(2022\)](#). Standard errors, reported in parentheses, are obtained by bootstrapping posterior estimates of detective-level bias obtained using [Morris \(1983\)](#)'s Empirical Bayes approach.

Table 1: Characteristics of the Victims, by Race

	<i>Homicide</i>			<i>Non-Fatal Violent Crime</i>		
	<i>White Victims</i>	<i>Black Victims</i>	<i>Hispanic Victims</i>	<i>White Victims</i>	<i>Black Victims</i>	<i>Hispanic Victims</i>
<i>Incident/Victim Information:</i>						
Female Victim	0.241	0.108	0.103	0.355	0.403	0.292
Victim's Age	39.046	29.453	26.917	35.753	32.126	30.679
Gang Related	0.123	0.144	0.383	0.018	0.022	0.098
Domestic Violence	0.137	0.062	0.055	0.075	0.219	0.118
Firearm	0.530	0.854	0.828	0.255	0.362	0.345
Assault				0.161	0.205	0.198
Battery				0.196	0.371	0.284
Sexual Assault				0.072	0.054	0.056
Robbery				0.571	0.371	0.462
<i>Status Information:</i>						
Cleared	0.660	0.450	0.453	0.325	0.332	0.306
Exceptional Clearance	0.156	0.139	0.115	0.115	0.154	0.110
Arrest and Prosecution	0.504	0.311	0.337	0.210	0.178	0.196
Victim Refuse to Cooperate				0.065	0.108	0.066
<i>Detectives Information:</i>						
Detective Assigned	0.988	0.998	0.996	0.881	0.868	0.875
Black Detective	0.062	0.123	0.067	0.081	0.211	0.076
Hispanic Detective	0.133	0.122	0.148	0.154	0.126	0.214
White Detective	0.780	0.742	0.766	0.733	0.644	0.679
Total Victims	618	9,792	2,273	102,165	477,724	154,141

Notes: This table shows the average characteristics of the victims of violent crime, with related information about the incident, investigation and assigned detective. The information is shown separately for White, Black and Hispanic victims and for homicides and nonfatal violent crime, and covers the time period 2001-2023. Only White, Black and Hispanic victims are kept in the sample.

Table 2: Characteristics of the Detectives

White Detective	0.658
Black Detective	0.157
Hispanic Detective	0.162
Native/AAPI Detective	0.023
Female Detective	0.189
Cases per Year	24.252
Share of Modal Crime Assigned	0.837
Specialist (Modal $\geq 75\%$)	0.723
Specialist (Modal $\geq 80\%$)	0.662
Specialist (Modal $\geq 90\%$)	0.494
(Mostly) Homicide Detective	0.036
(Mostly) Sex Assault Detective	0.084
(Mostly) Robbery Detective	0.304
(Mostly) Assault/Battery Detective	0.577
Detectives	3261

Notes: This table shows the average characteristics of the detectives in the sample. The dataset is at the detective level. Cases involving victims of all races are used for this analysis. The data covers the time period 2001-2023.

Table 3: Disparities in Clearance Probability

	(1) Cleared	(2) Cleared	(3) Cleared - No Victim Refusal	(4) Cleared - No Victim Refusal	(5) Cleared - Arrest and Prosecution	(6) Cleared - Arrest and Prosecution
Black Victim	-0.032*** (0.002)	-0.030*** (0.002)	-0.043*** (0.002)	-0.042*** (0.002)	-0.038*** (0.002)	-0.038*** (0.002)
Hispanic Victim	-0.027*** (0.002)	-0.024*** (0.002)	-0.020*** (0.002)	-0.019*** (0.002)	-0.013*** (0.002)	-0.013*** (0.002)
Pct Black (25%,50%]	0.000 (0.003)		-0.002 (0.003)		-0.002 (0.002)	
Pct Black (50%-75%]	-0.011*** (0.004)		-0.011*** (0.003)		-0.007*** (0.003)	
Pct Black (75%-100%]	-0.029*** (0.003)		-0.028*** (0.003)		-0.023*** (0.003)	
Pct Hispanic (25%,50%]	-0.008*** (0.003)		-0.008*** (0.003)		-0.005** (0.002)	
Pct Hispanic (50%-75%]	-0.020*** (0.003)		-0.018*** (0.003)		-0.012*** (0.003)	
Pct Hispanic (75%-100%]	-0.032*** (0.004)		-0.025*** (0.003)		-0.017*** (0.003)	
Ln(Median Per Capita Income)	0.001 (0.002)		0.005*** (0.002)		0.005*** (0.001)	
Area $\times$ Time $\times$ Crime Type FE	✓	✓	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓	✓	✓
Victim Gender and Age	✓	✓	✓	✓	✓	✓
Difficulty Controls	✓	✓	✓	✓	✓	✓
Tract FE		✓		✓		✓
White Victim Mean	0.2776	0.2776	0.2179	0.2179	0.1657	0.1657
Observations	643,997	643,997	643,997	643,997	643,997	643,997
R ²	0.1851	0.1873	0.1248	0.1270	0.1071	0.1095

Notes: This table shows the results from the estimation of Model 1. The outcome in Columns 1 and 2 is overall clearance; the outcome in Columns 3 and 4 is an indicator for a clearance not achieved due to the victim's refusal; the outcome in Columns 5 and 6 is an indicator for a clearance achieved explicitly via arrest and prosecution. Case controls include whether the crime is gang-related, whether the crime is domestic violence-related, fixed effects for the weapon used in the crime, fixed effects for the premises where the crime occurred (e.g. street or residence), whether it was daylight or dark at the time of the crime, the day of the week when the crime occurred, the initial priority of the 911 call when the crime occurred, and how the crime was reported. Each observation is weighted by the inverse of the total number of victims involved in the incident. Standard errors are clustered at the detective area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Gap in Acceptance Rate by Race of the Victim

	(1)	(2)	(3)	(4)
	FRU Approves	FRU Approves	FRU Approves	FRU Approves
Black Victim	-0.091*** (0.005)	-0.074*** (0.005)	-0.070*** (0.006)	-0.064*** (0.006)
Hispanic Victim	-0.035*** (0.006)	-0.018*** (0.005)	-0.019*** (0.006)	-0.022*** (0.006)
Crime Type FE		✓		
Area $\times$ Review Month-Year $\times$ Crime Type FE			✓	✓
Detective FE				✓
White Victim Mean	0.9084	0.9084	0.9084	0.9084
Observations	41,485	41,485	41,485	41,485
R ²	0.0103	0.0470	0.1883	0.2647

Notes: This table shows the results from the estimation of Equation 4. The dependent variable is a dummy that takes value 1 if the case is accepted by the FRU. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Columns 1 presents the raw disparity between victims of different races. Column 2 adds fixed effects for the type of violent crime the victim has experienced. Column 3 interacts crime fixed effects with area-by-time fixed effects. Column 4 adds detective fixed effects. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Adding Non-Race Characteristics of the Victim and the Incident

	(1)	(2)	(3)	(4)	(5)	(6)
	FRU Approves	FRU Approves	FRU Approves	FRU Approves	FRU Approves	FRU Approves
Black Victim	-0.064*** (0.006)	-0.064*** (0.006)	-0.057*** (0.007)	-0.056*** (0.007)	-0.056*** (0.007)	-0.055*** (0.007)
Hispanic Victim	-0.022*** (0.006)	-0.022*** (0.006)	-0.016** (0.006)	-0.017** (0.007)	-0.016** (0.007)	-0.015** (0.007)
Pct Black (25%,50%]			-0.010 (0.010)			
Pct Black (50%-75%]			0.002 (0.013)			
Pct Black (75%-100%]			-0.004 (0.012)			
Pct Hispanic (25%,50%]			0.001 (0.010)			
Pct Hispanic (50%-75%]			-0.006 (0.012)			
Pct Hispanic (75%-100%]			-0.003 (0.013)			
Ln(Median Per Capita Income)			0.014** (0.007)			
Black Arrestee						0.011 (0.011)
Hispanic Arrestee						0.005 (0.011)
Area $\times$ Review Month-Year $\times$ Crime Type FE	✓	✓	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓	✓	✓
Victim Gender and Age		✓	✓	✓	✓	✓
Tract FE				✓	✓	✓
Difficulty Controls					✓	✓
Arrestee Gender and Age						✓
White Victim Mean	0.9084	0.9084	0.9084	0.9084	0.9084	0.9084
Observations	41,485	41,485	41,449	41,485	41,485	41,485
R ²	0.2647	0.2659	0.2664	0.2888	0.2989	0.3022

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the case is accepted by the FRU. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Column 1 presents the baseline estimate without any additional controls. Difficulty controls include whether the crime is gang-related, whether the crime is domestic violence-related, fixed effects for the weapon used in the crime, fixed effects for the premises where the crime occurred (e.g. street or residence), whether it was daylight or dark at the time of the crime, the day of the week when the crime occurred, the initial priority of the 911 call when the crime occurred, and how the crime was reported. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Addressing Potential Infra-Marginality

	Controlling for Days		Days ≥ 1		Days ≥ 2		Days ≥ 7	
	(1) FRU Approves	(2) FRU Approves	(3) FRU Approves	(4) FRU Approves	(5) FRU Approves	(6) FRU Approves	(7) FRU Approves	(8) FRU Approves
Black Victim	-0.063*** (0.006)	-0.055*** (0.007)	-0.055*** (0.007)	-0.047*** (0.008)	-0.054*** (0.009)	-0.045*** (0.011)	-0.057*** (0.011)	-0.053*** (0.012)
Hispanic Victim	-0.022*** (0.006)	-0.015** (0.007)	-0.022*** (0.007)	-0.015* (0.008)	-0.024*** (0.009)	-0.013 (0.010)	-0.020* (0.011)	-0.001 (0.014)
Days to Review	-0.000*** (0.000)	-0.000*** (0.000)						
Days to Review ²	0.000*** (0.000)	0.000*** (0.000)						
Days to Review ³	-0.000** (0.000)	-0.000** (0.000)						
Black Arrestee		0.014 (0.011)		0.030** (0.013)		0.044*** (0.017)		0.035* (0.021)
Hispanic Arrestee		0.007 (0.011)		0.019 (0.014)		0.033* (0.018)		0.013 (0.022)
Area $\times$ Review Month-Year $\times$ Crime Type FE	✓	✓	✓	✓	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓	✓	✓	✓	✓
Victim Gender and Age		✓		✓		✓		✓
Arrestee Gender and Age		✓		✓		✓	✓	✓
Tract FE		✓		✓		✓		✓
Difficulty Controls		✓		✓		✓		✓
Observations	41,172	41,172	32,101	32,093	21,752	21,735	15,071	15,054
R ²	0.2660	0.3035	0.3052	0.3482	0.3848	0.4408	0.4621	0.5298

Notes: This table shows the results from the estimation of Equation 4 with additional controls and sample restrictions. The dependent variable is a dummy that takes value 1 if the case is accepted by the FRU. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Guilt as Outcome

	(1) Guilty	(2) Guilty	(3) Guilty	(4) Guilty
Black Victim	-0.093*** (0.010)	-0.077*** (0.011)	-0.077*** (0.011)	-0.076*** (0.011)
Hispanic Victim	-0.030*** (0.010)	-0.016 (0.011)	-0.016 (0.011)	-0.014 (0.011)
Number of Charges	0.000** (0.000)	0.000* (0.000)	0.000** (0.000)	0.000** (0.000)
Pct Black (25%,50%]		-0.011 (0.016)		
Pct Black (50%-75%]		-0.014 (0.020)		
Pct Black (75%-100%]		-0.030 (0.019)		
Pct Hispanic (25%,50%]		-0.003 (0.016)		
Pct Hispanic (50%-75%]		-0.017 (0.018)		
Pct Hispanic (75%-100%]		-0.031 (0.021)		
Ln(Median Per Capita Income)		0.009 (0.011)		
Black Arrestee				0.013 (0.017)
Hispanic Arrestee				0.003 (0.018)
Area $\times$ Review Month-Year $\times$ Crime Type FE	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓
Victim Gender and Age		✓	✓	✓
Arrestee Gender and Age		✓	✓	✓
Tract FE			✓	✓
Difficulty Controls		✓	✓	✓
White Victim Mean	0.7849	0.7849	0.7849	0.7849
Observations	28,657	28,641	28,651	28,651
R ²	0.2265	0.2373	0.2733	0.2757

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the case is linked with any guilty ruling in court. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Recidivism

	(1)	(2)	(3)	(4)
	3 Years Violent Recidivism	3 Years Violent and Felony Recidivism	1 Year Violent Recidivism	1 Year Violent and Felony Recidivism
Black Victim	0.050*** (0.010)	0.038*** (0.010)	0.028*** (0.009)	0.020** (0.008)
Hispanic Victim	0.031*** (0.010)	0.027*** (0.010)	0.010 (0.008)	0.008 (0.008)
Black Arrestee	0.062*** (0.015)	0.063*** (0.014)	0.032** (0.013)	0.035*** (0.012)
Hispanic Arrestee	0.023 (0.015)	0.024* (0.014)	0.015 (0.013)	0.014 (0.012)
Area $\times$ Review Month-Year $\times$ Crime Type FE	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓
Victim Gender and Age	✓	✓	✓	✓
Arrestee Gender and Age	✓	✓	✓	✓
Tract FE	✓	✓	✓	✓
Difficulty Controls	✓	✓	✓	✓
White Victim Mean	0.2947	0.2758	0.1587	0.1441
Observations	38,018	38,018	38,018	38,018
R ²	0.2569	0.2472	0.2233	0.2172

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the arrestee is take in into custody for another index violent crime in the future in Columns 1 and 3, and for index violent crime that also involved a felony in Columns 2 and 4. Columns 1 and 2 consider a three year horzion, while Columns 3 and 4 consider a one year horizon. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

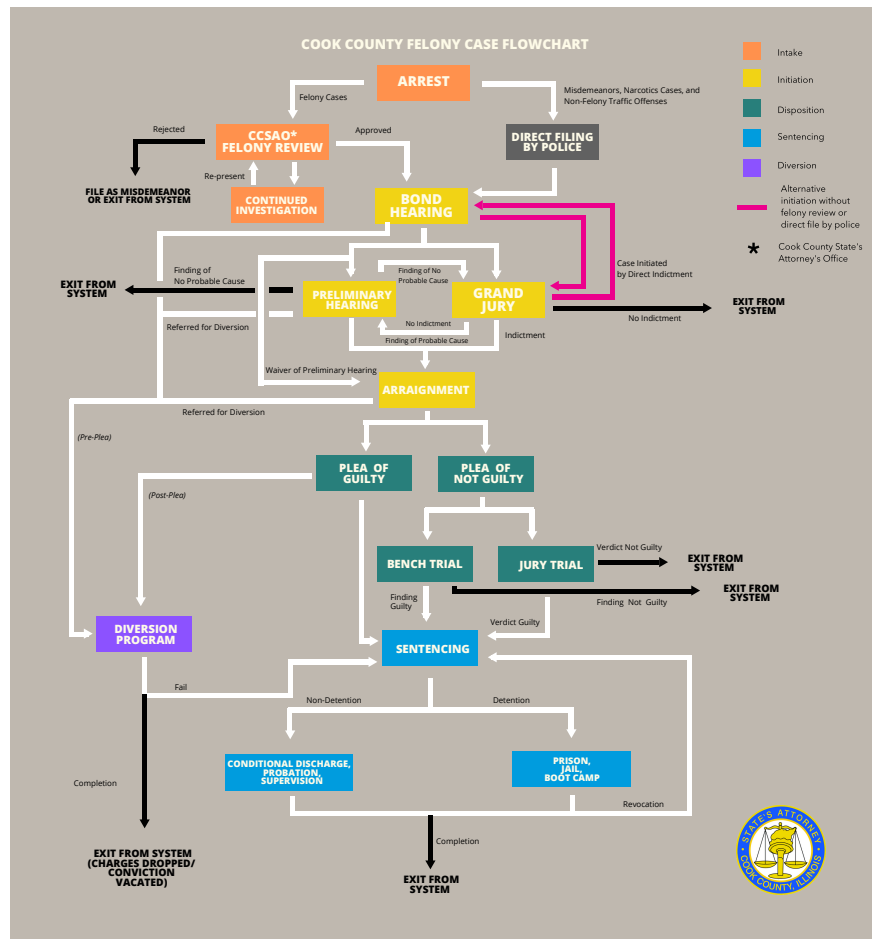
Table 9: Heterogeneity by Detective's Race and Experience at Clearance

	(1)	(2)
	FRU Approves	FRU Approves
Victim and Detective Race Match	-0.003 (0.007)	
Years of Experience at Review	0.012* (0.007)	0.012* (0.007)
Black Victim $\times$ Years of Experience at Review		-0.000 (0.001)
Hispanic Victim $\times$ Years of Experience at Review		0.000 (0.001)
Area $\times$ Review Month-Year $\times$ Crime Type FE	✓	✓
Detective FE	✓	✓
Tract FE	✓	✓
Difficulty Controls	✓	✓
Victim Race, Gender and Age	✓	✓
Defendant Race, Gender and Age	✓	✓
Observations	41,259	41,259
R ²	0.3023	0.3023

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the case is accepted by the FRU. The main regressors of interest in Column 1 are the interaction between dummies for the race of the victim and the experience of the detective at the time of review and the baseline coefficient for experience. In Column 2 the main regressor of interest is a dummy that takes value 1 if the race of the victim and the race of the detective match. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix A: Supplemental Figures and Tables

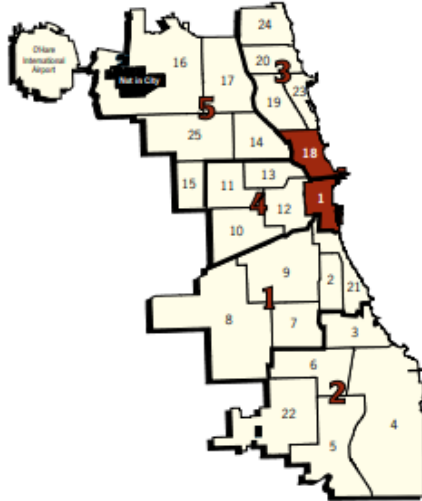
Figure A.1: The Arrest-to-Court Pipeline in Cook County, IL



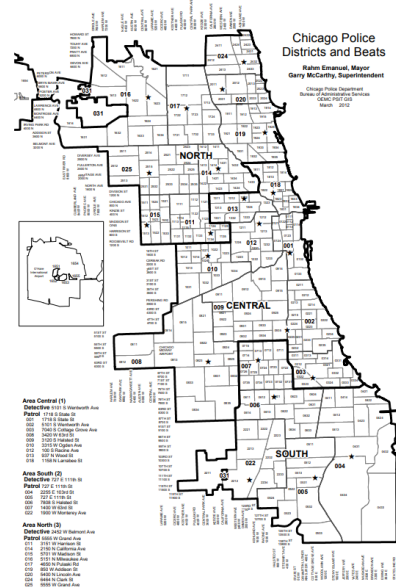
Notes: This figure shows the process that goes from entry to exit in the criminal justice system in Cook County, IL.
Source: Cook County State Attorney's Office.

Figure A.2: Map of the City of Chicago with Detective Areas and Districts Labeled

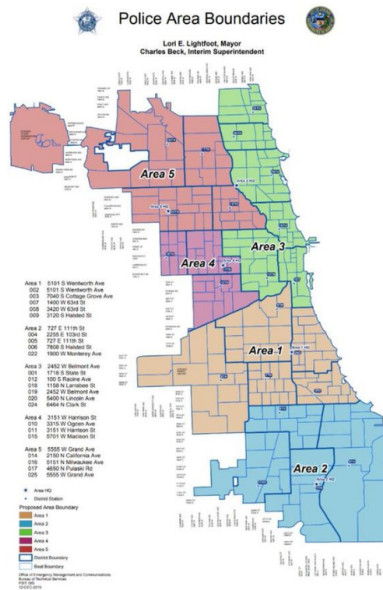
Chicago Police Areas and Districts



(a) Before Closures in 3/2012



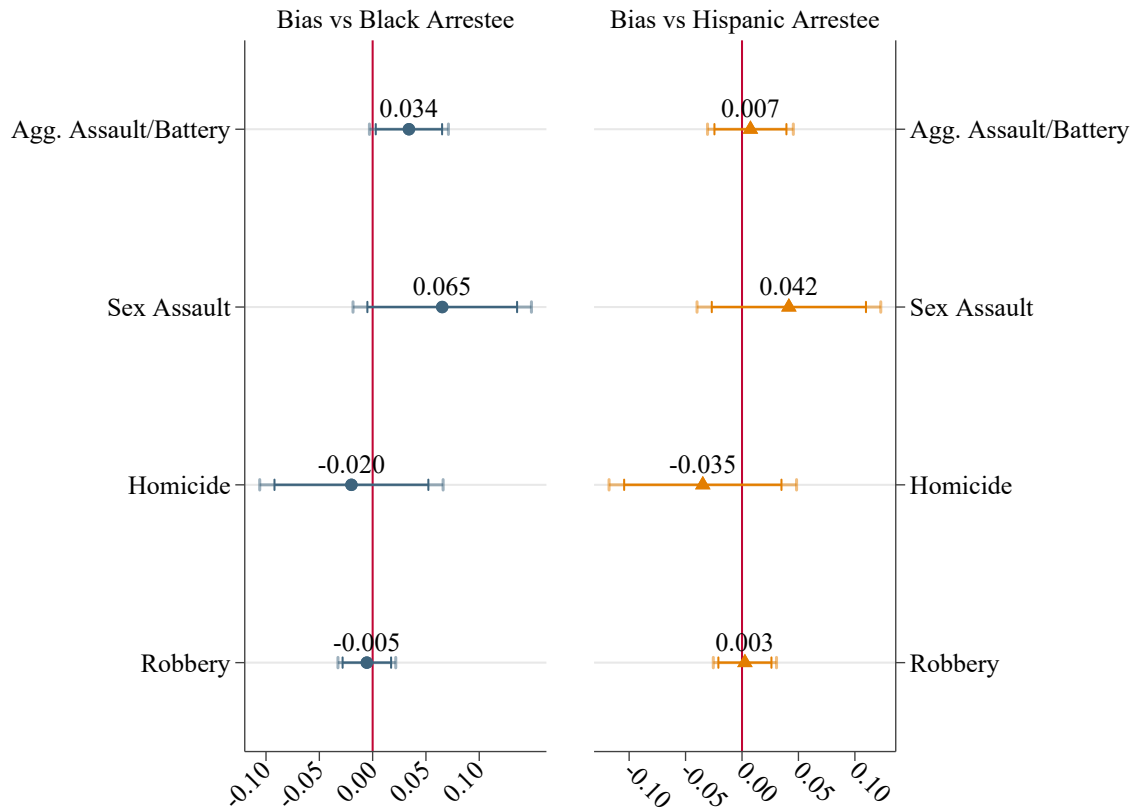
(b) After Closures in 3/2012 and Before Re-openings in 4/2020



(c) After Reopenings in 4/2020

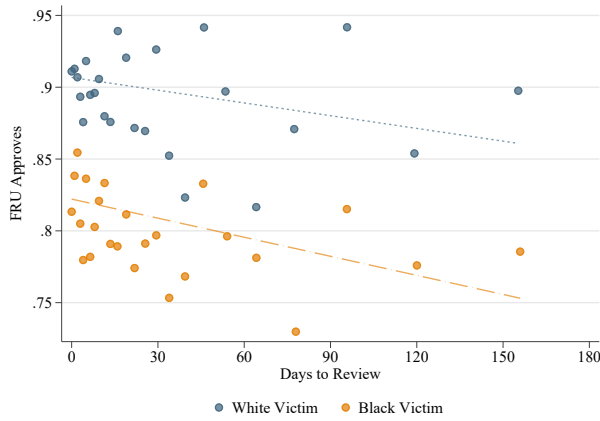
Notes: This figure shows the evolution of the detective areas' boundaries over time in Chicago, before closures in 3/2012 (Panel a), after closures in 3/2012 and before reopenings in 4/2020, and after reopenings in 2/2020

Figure A.3: Heterogeneity in Racial Bias by Type of Violent Crime

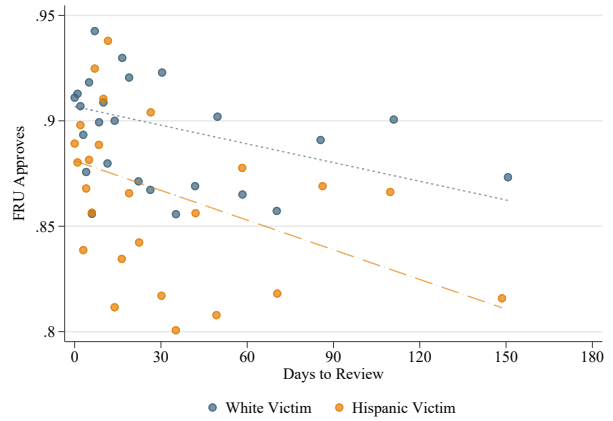


Notes: This figure plots the coefficient from a heterogeneous version of the model used in Column 6 of 5 by type of violent crime. Race dummies for victims and arrestees are interacted with fixed effects for the type of violent crime. This figure shows the coefficients on Black and Hispanic arrestees, separately.

Figure A.4: Relationship between Acceptance and Days to Clearance



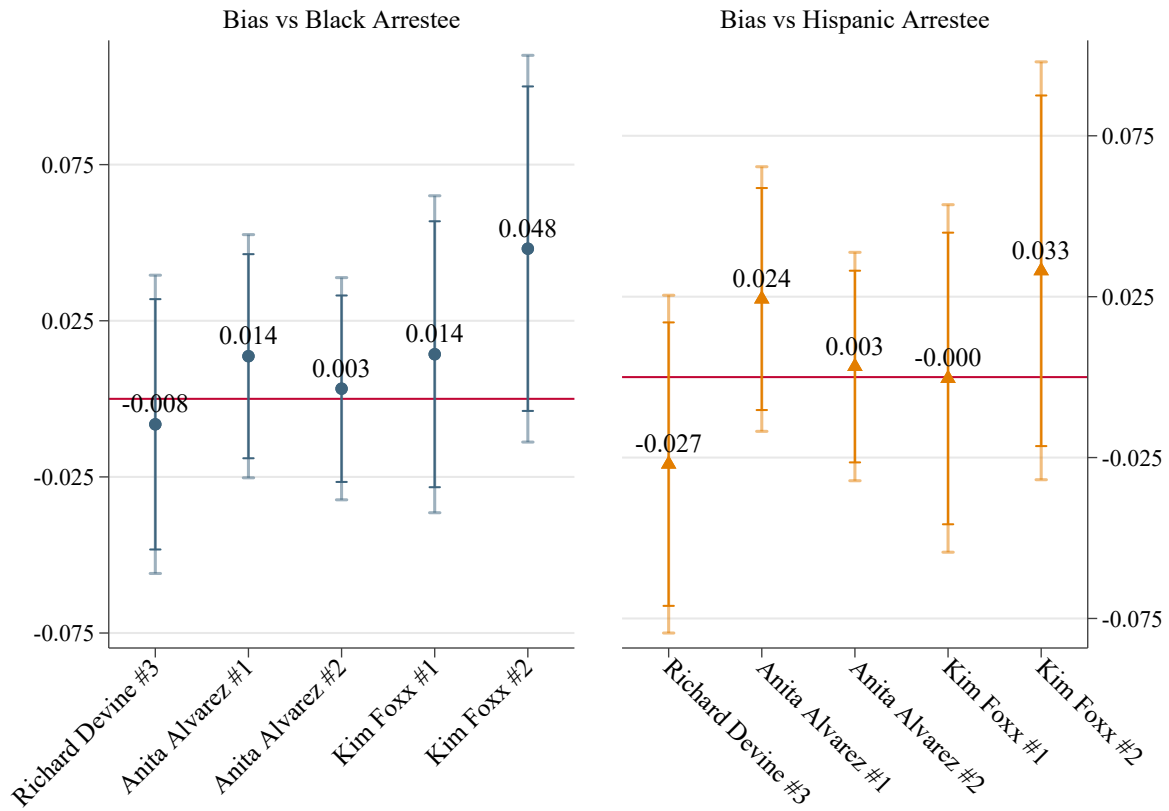
(a) White Victim vs. Black Victim



(b) White Victim vs. Hispanic Victim

Notes: This figure plots a bin scatter of the relationship between acceptance by FRU and days elapsed from report of the crime to review, by race of the victim. The range of days to review is truncated at 180 days.

Figure A.5: Heterogeneity in Racial Bias by SA's Term



Notes: This figure plots the coefficient from a heterogeneous version of the model used in Column 6 of 5 by term of the SA leading the CCSAO. Race dummies for victims and arrestees are interacted with fixed effects for the term of the SA leading the CCSAO. This figure shows the coefficients on Black and Hispanic arrestees, separately.

Table A.1: Correlation Between Victim and Detective Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Black Det.	Hispanic Det.	White Det.	Leave-Out Cl. Rate	Leave-Out Arrest Cl. Rate	Black Det.	Hispanic Det.	White Det.	Leave-Out Cl. Rate	Leave-Out Arrest Cl. Rate
Black Victim	0.052*** (0.009)	-0.010 (0.011)	-0.038*** (0.013)	0.004*** (0.001)	-0.002*** (0.001)	0.017*** (0.003)	-0.002 (0.003)	-0.016*** (0.004)	-0.002*** (0.001)	-0.001*** (0.000)
Hispanic Victim	-0.005 (0.004)	0.035*** (0.010)	-0.030*** (0.011)	-0.003** (0.001)	-0.000 (0.001)	0.003* (0.002)	0.019*** (0.004)	-0.021*** (0.004)	-0.002*** (0.001)	-0.000 (0.000)
Area $\times$ Time $\times$ Crime Type FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Case Controls										
Observations	619645	619645	619645	619645	619645	619645	619645	619645	619645	619645

Notes: This table shows the correlation between victim race and the characteristics of the detective first assigned to the case. The regressions condition on conditional on area-by-time-by-crime type fixed effects, which is the level of assignment, and Columns 6 to 10 add case controls as well. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. The dependent variables are the race of the detective, a leave-one-out measure of clearance rate and a leave-one-out measure of arrest clearance rate. Standard errors are clustered at the detective level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.2: Disparities in Clearance Probability for crimes reported from 2006 onward

	(1) Cleared	(2) Cleared	(3) Cleared - No Victim Refusal	(4) Cleared - No Victim Refusal	(5) Cleared - Arrest and Prosecution	(6) Cleared - Arrest and Prosecution
Black Victim	-0.035*** (0.003)	-0.032*** (0.003)	-0.046*** (0.002)	-0.045*** (0.002)	-0.038*** (0.002)	-0.038*** (0.002)
Hispanic Victim	-0.027*** (0.003)	-0.025*** (0.003)	-0.021*** (0.002)	-0.020*** (0.002)	-0.014*** (0.002)	-0.013*** (0.002)
Pct Black (25%,50%]	0.000 (0.004)		-0.002 (0.003)		-0.002 (0.003)	
Pct Black (50%-75%]	-0.016*** (0.004)		-0.015*** (0.004)		-0.011*** (0.003)	
Pct Black (75%-100%]	-0.031*** (0.004)		-0.030*** (0.004)		-0.023*** (0.003)	
Pct Hispanic (25%,50%]	-0.007** (0.003)		-0.007** (0.003)		-0.003 (0.003)	
Pct Hispanic (50%-75%]	-0.022*** (0.004)		-0.018*** (0.004)		-0.012*** (0.003)	
Pct Hispanic (75%-100%]	-0.033*** (0.005)		-0.028*** (0.004)		-0.018*** (0.003)	
Ln(Median Per Capita Income)	0.003 (0.002)		0.006*** (0.002)		0.006*** (0.002)	
Area $\times$ Time $\times$ Crime Type FE	✓	✓	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓	✓	✓
Victim Gender and Age	✓	✓	✓	✓	✓	✓
Difficulty Controls	✓	✓	✓	✓	✓	✓
Tract FE		✓		✓		✓
White Victim Mean	0.2788	0.2788	0.2193	0.2193	0.1644	0.1644
Observations	461,951	461,951	461,951	461,951	461,951	461,951
R ²	0.1817	0.1844	0.1205	0.1233	0.1052	0.1081

Notes: This table shows the results from the estimation of Model 1. The outcome in Columns 1 and 2 is overall clearance; the outcome in Columns 3 and 4 is an indicator for a clearance not achieved due to the victim's refusal; the outcome in Columns 5 and 6 is an indicator for a clearance achieved explicitly via arrest and prosecution. Case controls include whether the crime is gang-related, whether the crime is domestic violence-related, fixed effects for the weapon used in the crime, fixed effects for the premises where the crime occurred (e.g. street or residence), whether it was daylight or dark at the time of the crime, the day of the week when the crime occurred, the initial priority of the 911 call when the crime occurred, and how the crime was reported. Each observation is weighted by the inverse of the total number of victims involved in the incident. Standard errors are clustered at the detective area-time-crime type level. The sample is restricted to crime reported from 2006 afterwards. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3: Characteristics of the Sample for the Outcome Test

	<i>Homicide</i>			<i>Non-Fatal Violent Crime</i>		
	<i>White Victims</i>	<i>Black Victims</i>	<i>Hispanic Victims</i>	<i>White Victims</i>	<i>Black Victims</i>	<i>Hispanic Victims</i>
Female Victim	0.264	0.164	0.161	0.398	0.426	0.408
Victim's Age	40.597	31.005	27.123	35.883	31.489	29.144
Black Suspect	0.406	0.968	0.215	0.660	0.960	0.405
Hispanic Suspect	0.294	0.031	0.711	0.174	0.029	0.543
White Suspect	0.320	0.005	0.098	0.177	0.013	0.064
Female Suspect	0.069	0.102	0.058	0.075	0.099	0.066
Assault				0.032	0.079	0.060
Battery				0.219	0.336	0.218
Robbery				0.653	0.443	0.547
Sexual Assault				0.096	0.143	0.175
FRU Approves	0.872	0.820	0.832	0.905	0.811	0.872
Total Observations	230	2,368	574	6,456	16,815	8,595

Notes: This table shows the average characteristics of the victims of violent crime, with related information about the incident, arrestee, investigation and assigned detective, for those observations that enter the sample of the outcome test. The information is shown separately for White, Black and Hispanic victims and for homicides and nonfatal violent crime. The sample includes offenses reported between 2001 and 2023 and arrestees taken into custody from 2006 and 2023 and charged by the police for at least one felony. Only White, Black and Hispanic victims, and White, Black and Hispanic arrestees are kept in the sample.

Table A.4: Top Charge Guilty

	(1)	(2)	(3)	(4)
	Top Charge Guilty	Top Charge Guilty	Top Charge Guilty	Top Charge Guilty
Black Victim	-0.122*** (0.010)	-0.094*** (0.012)	-0.090*** (0.012)	-0.091*** (0.012)
Hispanic Victim	-0.060*** (0.011)	-0.033*** (0.012)	-0.025** (0.012)	-0.023* (0.012)
Number of Charges	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
Pct Black (25%,50%]		-0.009 (0.016)		
Pct Black (50%-75%]		-0.028 (0.022)		
Pct Black (75%-100%]		-0.027 (0.021)		
Pct Hispanic (25%,50%]		-0.012 (0.017)		
Pct Hispanic (50%-75%]		-0.025 (0.020)		
Pct Hispanic (75%-100%]		-0.030 (0.023)		
Ln(Median Per Capita Income)		0.021* (0.012)		
Black Arrestee				0.043** (0.019)
Hispanic Arrestee				0.017 (0.020)
Area $\times$ Review Month-Year $\times$ Crime Type FE	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓
Victim Gender and Age		✓	✓	✓
Arrestee Gender and Age		✓	✓	✓
Tract FE			✓	✓
Difficulty Controls		✓	✓	✓
White Victim Mean	0.6472	0.6472	0.6472	0.6472
Observations	28,657	28,641	28,651	28,651
R ²	0.2682	0.2774	0.3130	0.3168

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the top charge brought against the arrestee is associated with a guilty ruling. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Appears at Arraignment

	(1) Arraignment	(2) Arraignment	(3) Arraignment	(4) Arraignment
Black Victim	-0.081*** (0.009)	-0.074*** (0.010)	-0.075*** (0.010)	-0.074*** (0.010)
Hispanic Victim	-0.025*** (0.009)	-0.017* (0.009)	-0.021** (0.009)	-0.021** (0.010)
Number of Charges	0.000** (0.000)	0.000* (0.000)	0.000** (0.000)	0.000** (0.000)
Pct Black (25%,50%]		-0.018 (0.014)		
Pct Black (50%-75%]		-0.003 (0.018)		
Pct Black (75%-100%]		-0.017 (0.017)		
Pct Hispanic (25%,50%]		0.006 (0.013)		
Pct Hispanic (50%-75%]		-0.005 (0.016)		
Pct Hispanic (75%-100%]		-0.014 (0.018)		
Ln(Median Per Capita Income)		0.011 (0.010)		
Black Arrestee				0.008 (0.015)
Hispanic Arrestee				0.005 (0.015)
Area $\times$ Review Month-Year $\times$ Crime Type FE	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓
Victim Gender and Age		✓	✓	✓
Arrestee Gender and Age		✓	✓	✓
Tract FE			✓	✓
Difficulty Controls		✓	✓	✓
White Victim Mean	0.8562	0.8562	0.8562	0.8562
Observations	28,657	28,641	28,651	28,651
R ²	0.2352	0.2494	0.2829	0.2857

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the arrestee appears at arraignment. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Status of the Investigation

	Full Sample		1 Arrestee Involved	
	(1) FRU Approves	(2) Cleared - Arrest and Prosecution	(3) FRU Approves	(4) Cleared - Arrest and Prosecution
Black Victim	-0.055*** (0.007)	-0.025*** (0.007)	-0.058*** (0.008)	-0.027*** (0.009)
Hispanic Victim	-0.015** (0.007)	-0.010 (0.007)	-0.010 (0.008)	-0.005 (0.009)
Black Arrestee	0.011 (0.011)	-0.012 (0.011)	0.025* (0.013)	-0.002 (0.013)
Hispanic Arrestee	0.005 (0.011)	0.013 (0.012)	0.005 (0.013)	0.019 (0.013)
Area $\times$ Review Month-Year $\times$ Crime Type FE	✓	✓	✓	✓
Victim Gender and Age	✓	✓	✓	✓
Arrestee Gender and Age	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓
Tract FE	✓	✓	✓	✓
Difficulty Controls	✓	✓	✓	✓
White Victim Mean	0.9084	0.8837	0.9084	0.8837
Observations	41,485	41,485	28,049	28,049
R ²	0.3022	0.2910	0.3435	0.3234

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the status of the investigation is clearance by arrest and prosecution. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.7: Recidivism

	(1)	(2)	(3)	(4)
	3 Years Violent Recidivism	3 Years Violent and Felony Recidivism	1 Year Violent Recidivism	1 Year Violent and Felony Recidivism
Black Victim	0.048*** (0.010)	0.036*** (0.010)	0.029*** (0.009)	0.020** (0.008)
Hispanic Victim	0.030*** (0.010)	0.026*** (0.010)	0.010 (0.008)	0.008 (0.008)
Black Arrestee	0.059*** (0.015)	0.060*** (0.015)	0.033** (0.013)	0.035*** (0.013)
Hispanic Arrestee	0.016 (0.015)	0.017 (0.015)	0.013 (0.013)	0.011 (0.012)
Pct Black (25%,50%]	0.002 (0.014)	-0.002 (0.014)	-0.003 (0.012)	-0.008 (0.011)
Pct Black (50%-75%]	0.004 (0.018)	-0.003 (0.018)	-0.017 (0.015)	-0.021 (0.014)
Pct Black (75%-100%]	0.020 (0.017)	0.008 (0.016)	-0.001 (0.014)	-0.008 (0.014)
Pct Hispanic (25%,50%]	0.011 (0.014)	0.008 (0.013)	0.007 (0.011)	0.002 (0.011)
Pct Hispanic (50%-75%]	0.008 (0.015)	-0.002 (0.015)	0.015 (0.012)	0.011 (0.012)
Pct Hispanic (75%-100%]	0.011 (0.018)	0.009 (0.018)	0.018 (0.015)	0.017 (0.014)
Ln(Median Per Capita Income)	-0.001 (0.010)	-0.001 (0.009)	0.008 (0.008)	0.007 (0.008)
Area $\times$ Review Month-Year $\times$ Crime Type FE	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓
Victim Gender and Age	✓	✓	✓	✓
Arrestee Gender and Age	✓	✓	✓	✓
Difficulty Controls	✓	✓	✓	✓
White Victim Mean	0.2947	0.2758	0.1587	0.1441
Observations	37,988	37,988	37,988	37,988
R ²	0.2304	0.2208	0.1962	0.1909

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the arrestee is take in into custody for another index violent crime in the future in Columns 1 and 3, and for index violent crime that also involved a felony in Columns 2 and 4. Columns 1 and 2 consider a three year horizon, while Columns 3 and 4 consider a one year horizon. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Additional Robustness Checks

	1 Victim	1 Arrestee	1 Victim and 1 Arrestee	No Weighting	Detective Clustering	First Detective
	(1)	(2)	(3)	(4)	(5)	(6)
	FRU Approves	FRU Approves	FRU Approves	FRU Approves	FRU Approves	FRU Approves
Black Victim	-0.057*** (0.008)	-0.058*** (0.008)	-0.064*** (0.010)	-0.050*** (0.007)	-0.055*** (0.007)	-0.056*** (0.007)
Hispanic Victim	-0.020** (0.008)	-0.010 (0.008)	-0.015 (0.010)	-0.016** (0.006)	-0.015** (0.006)	-0.015** (0.007)
Black Arrestee	0.016 (0.012)	0.025* (0.013)	0.033** (0.015)	0.000 (0.010)	0.011 (0.010)	0.015 (0.011)
Hispanic Arrestee	0.007 (0.013)	0.005 (0.013)	0.009 (0.015)	0.002 (0.011)	0.005 (0.011)	0.009 (0.011)
Area $\times$ Review Month-Year $\times$ Crime Type FE	✓	✓	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓	✓	✓
Victim Gender and Age	✓	✓	✓	✓	✓	✓
Arrestee Gender and Age	✓	✓	✓	✓	✓	✓
Tract FE	✓	✓	✓	✓	✓	✓
Difficulty Controls	✓	✓	✓	✓	✓	✓
White Victim Mean	0.9084	0.9084	0.9084	0.9084	0.9084	0.9093
Observations	27,945	28,049	19,897	41,485	41,485	41,578
R ²	0.3359	0.3435	0.3764	0.3006	0.3022	0.2976

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the status of the investigation is clearance by arrest and prosecution. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.9: Estimates in the Sample used to Estimate Detective Level Bias

	(1) FRU Approves	(2) FRU Approves
Black Victim	-0.063*** (0.010)	-0.040*** (0.013)
Hispanic Victim	-0.014 (0.009)	-0.008 (0.010)
Black Arrestee		0.020 (0.017)
Hispanic Arrestee		0.030* (0.017)
Area $\times$ Review Month-Year $\times$ Crime Type FE	✓	✓
Detective FE	✓	✓
Victim Gender and Age		✓
Arrestee Gender and Age		✓
Tract FE		✓
Difficulty Controls		✓
White Victim Mean	0.9084	0.9084
Observations	11,512	11,464
R ²	0.2543	0.3491

Notes: This table shows the results from the estimation of Equation 4 in the sample to estimate detective level bias as displayed in Figure 6. Column 1 uses the same model used in Column 4 of Table 4. Column 2 uses the same model used in Column 6 of Table 5. The dependent variable is a dummy that takes value 1 if the case is accepted by the FRU. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix B: Theory Appendix

B.1 Proof of Proposition 1

We can show that if there is a gap in success rates between Black and White cases that are solved and handed over to the prosecutor after the same number of days t_i^* , the investigator displays racial bias:

$$\begin{aligned}
& E[Y_i \mid R_i = b, D_i = 1] - E[Y_i \mid R_i = w, D_i = 1] \\
&= E[Y_i^*(t) \mid R_i = b, T_i = t_i^*] - E[Y_i^*(t) \mid R_i = w, T_i = t_i^*] \\
&= E[Y_i^*(t) \mid R_i = b, p_i(t) = c(b)] - E[Y_i^*(t) \mid R_i = w, p_i(t) = c(w)] \\
&= E[E[Y_i^*(t) \mid R_i, U_i] \mid R_i = b, p_i(t) = c(b)] - \\
&\quad E[E[Y_i^*(t) \mid R_i, U_i] \mid R_i = w, p_i(t) = c(w)] \\
&= E[p_i(t) \mid R_i = b, p_i(t) = c(b)] - E[p_i(t) \mid R_i = w, p_i(t) = c(w)] \\
&= c(b) - c(w).
\end{aligned} \tag{8}$$

The first equality follows because $Y_i = Y_i^*(T_i)$ and $T_i = t_i^* \iff D_i = 1$. The second equality follows because $T_i = t_i^* \iff p_i(t_i^*) = c(r)$. The third equality follows by the law of iterated expectations. The fourth equality follows because $p_i(t) = E[Y_i^*(t) \mid R_i, U_i]$. The fifth equality follows by the conditioning on $p_i(t)$.

B.2 Microfoundation of the thresholds $c(R_i, t)$

A detective decides whether to bring the case of victim i to the DA's office, represented by $D_i \in \{0, 1\}$. If the case is submitted, this can either be successful if the DA accepts the charges against the suspect related to the case of victim i ($Y_i^* = 1$) or a failure if the DA rejects charges ($Y_i^* = 0$). In each period, the detective has beliefs $p_i(t)$ over the success of the case. Upon clearance, she receives utility κ regardless of the success status of the case. She dislikes making both false positives ($D_i = 1, Y_i^* = 0$) and false negatives ($D_i = 0, Y_i^* = 1$). Define the cost of these errors as β and γ , respectively. Her utility in period t is therefore:

$$U(t) = \kappa - \beta D_i(t)(1 - Y_i^*(t)) - \gamma(1 - D_i(t))Y_i^*(t), \tag{9}$$

with expected utility for different actions $d \in \{0, 1\}$ being

$$\begin{aligned}
E[U(t) \mid D_i(t) = 0] &= \kappa - \gamma p_i(t) \\
E[U(t) \mid D_i(t) = 1] &= \kappa - \beta(1 - p_i(t))
\end{aligned} \tag{10}$$

Her optimal decision rule is then

$$\begin{aligned}
D_i &= \mathbf{1}[-\beta(1 - p_i(t)) \geq -\gamma p_i(t)] \\
&= \mathbf{1}[p_i(t) \geq \frac{\beta(R_i)}{\beta(R_i) + \gamma(R_i)} = c(R_i)]
\end{aligned} \tag{11}$$

From the last equation, we see that $c(R_i)$ can be recast as the relative cost of type I error upon submission at period t of a case of victim of race R_i . We can therefore redefine racial bias, $c(b) <$

$c(w)$, as the detective having a lower relative cost of an unsuccessful submission when the victim is Black rather than White. This difference in relative costs can arise in several ways. The detective can perceive, for example, race-neutral costs of false negatives ($\gamma(b) = \gamma(w)$) but a lower cost of false positives for Black victims than for White victims ($\beta(b) < \beta(w)$): submitting a shoddy investigation at time t is less costly if the victim is Black rather than White. Alternatively, the detective can perceive, for example, race-neutral costs of false negatives ($\beta(b) = \beta(w)$) but a larger cost of false positives for Black victims than for White victims ($\gamma(b) > \gamma(w)$): the detective dislikes keeping potentially successful cases with Black victims open more than those of White victims since keeping a case open entails additional investigative effort, which under discrimination against Black victims is more costly for the latter group than for White victims. A combination of these two scenarios ($\beta(b) < \beta(w)$ and $\gamma(b) > \gamma(w)$) can also yield a lower relative cost of failure for cases with Black victims than for cases with White victims.